Tropical Medicine & International Health

SCOPING REVIEW

AI-Enhanced Point-of-Care Diagnostics for Infectious Diseases in Resource-Limited Settings: A Scoping Review

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ABSTRACT

Objectives: To systematically map the extent and nature of research on AI-enhanced point-of-care (POC) and rapid diagnostic technologies for infectious diseases in resource-limited settings, and to identify gaps in disease coverage, geographic representation and validation rigour.

Methods: This scoping review followed JBI methodology and PRISMA-ScR guidelines. The protocol was registered on OSF (https://doi.org/10.17605/OSF.IO/KV8MP). Five databases (PubMed, Embase, Scopus, Web of Science and IEEE Xplore) were searched for studies published from January 2015 to March 2026. Title/abstract and full-text screening used rule-based keyword screening with manual validation (Cohen's kappa = 0.856). Data were extracted using a 19-variable charting form and enriched with PubMed Central full texts.

Results: From 1072 records, 551 remained after deduplication and 237 studies were included. Publication volume grew exponentially, with 44% published in 2025–2026. COVID-19 (32%), malaria (27%) and tuberculosis (14%) dominated; neglected tropical diseases accounted for fewer than 8%. Microscopy (21%), molecular diagnostics (17%), biosensors (14%) and rapid diagnostic tests (14%) were the most common modalities. Convolutional neural networks predominated (26%), followed by random forests (10%) and support vector machines (8%). Only 7% of studies reported prospective field validation, while 62% did not report validation level. Geographic analysis revealed concentration in East Africa and South Asia, with underrepresentation of West Africa and Latin America.

Conclusions: AI-enhanced POC diagnostics for infectious diseases in resource-limited settings is a rapidly growing field facing critical gaps in validation rigour, disease equity and geographic representation. Only 16 of 237 studies (6.8%) report prospective field validation. Future research should prioritise field validation, expand beyond the COVID-19/malaria/TB triad and involve end-user communities from the design stage.

1 | Introduction

Point-of-care (POC) diagnostics are critical for infectious disease management in resource-limited settings, where centralised laboratory infrastructure is scarce, specialist clinicians are few and timely diagnosis directly affects treatment outcomes and transmission dynamics [1, 2]. The World Health Organization's (WHO) Essential Diagnostics List and universal health coverage framework underscore the urgency of expanding diagnostic access globally [3], yet a persistent 'diagnostic gap' remains: the disparity between available diagnostic technologies and their accessibility in low- and middle-income countries (LMICs) [4]. The 2021 Lancet Commission on diagnostics estimated that 47% of the global population has little or no access to diagnostics, with the burden falling disproportionately on sub-Saharan Africa and South Asia

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[4]. This gap has profound consequences: delayed or missed diagnoses contribute to inappropriate treatment, antimicrobial resistance, continued transmission and preventable mortality [4]. For infectious diseases in particular, the ability to diagnose rapidly at the point of care is essential not only for individual patient management but also for public health surveillance and outbreak response [2].

The diagnostic landscape in LMICs is shaped by well-documented structural constraints. Many health facilities in sub-Saharan Africa and South Asia lack reliable electricity, cold chain infrastructure, internet connectivity and trained laboratory personnel [5]. Community health workers, who serve as the front line of primary healthcare in many LMICs, are often tasked with interpreting diagnostic tests—such as malaria rapid diagnostic tests (RDTs) and HIV lateral flow assays—with minimal training and without access to quality assurance systems [6]. Reader variability, subjective interpretation of faint test lines and errors in blood film microscopy are well-documented sources of diagnostic inaccuracy in these settings [6]. The challenge extends beyond developing new technologies to ensuring that existing and emerging tools can be used accurately by the health workforce available at the point of care.

Artificial intelligence (AI) offers one route to narrowing this gap. AI-enhanced diagnostics can automate the interpretation of test results, improve accuracy in the hands of non-specialist health workers and enable quality assurance at scale [7, 8]. Applications span a broad range of modalities: from smartphone-based microscopy for malaria parasites [9] to AI-powered lateral flow assay readers for HIV and tuberculosis (TB) [10], algorithm-driven interpretation of chest radiographs [11] and deep learning analysis of point-of-care ultrasound images [12]. AI standardises interpretation across operators of varying skill levels, functions on low-cost hardware such as smartphones now ubiquitous in low-resource settings, and provides real-time decision support without specialist referral [7]. Global health organisations including the Global Fund, the Foundation for Innovative New Diagnostics (FIND) and PATH are actively investing in AI-powered diagnostic tools for deployment in resource-limited settings [13, 14]. The WHO has endorsed AI-assisted computer-aided detection for TB screening as part of its consolidated guidelines on tuberculosis, signalling institutional readiness for the integration of AI into diagnostic pathways in LMICs [15].

The field has grown rapidly. The convergence of three trends—the proliferation of low-cost smartphone hardware, advances in deep learning and computer vision and the COVID-19 pandemic's demand for decentralised testing—has driven a surge of research at the intersection of AI and POC diagnostics [16]. The pandemic served as a catalyst, generating unprecedented investment in rapid, decentralised diagnostic solutions and accelerating the development and deployment of AI-enhanced tools for antigen test interpretation, chest imaging analysis and cough-based screening algorithms [16]. Yet this rapid growth has been unevenly distributed across diseases, geographies and levels of validation. Most AI diagnostic tools remain at the proof-of-concept stage, with limited evidence of real-world deployment in the settings for which they are designed [17].

Existing reviews in this space are narrow in scope. Schwalbe and Wahl [18] addressed AI and global health broadly in *The Lancet*, with diagnostics as one of several application areas but without detailed mapping of POC-specific technologies. Wahl et al. [19] examined how AI could contribute to health in resource-poor settings, identifying the potential of diagnostic applications but not mapping the specific intersection of AI, POC testing and infectious diseases. A 2026 scoping review by von Bahr et al. [20] in *JMIR* covered AI-supported digital microscopy diagnostics in primary healthcare laboratories but was limited to a single modality. Reviews of AI-assisted TB screening [21] and AI for malaria microscopy [22] have addressed individual disease-modality combinations but not the broader landscape. The full intersection of AI, POC diagnostics, infectious disease and resource-limited settings remains unmapped, leaving policymakers, funders and researchers without a comprehensive evidence map to guide investment and research priorities.

The growth, dispersion and uneven validation noted above suggest a field that produces studies prolifically but converts them into deployable tools rarely. Mapping that pattern descriptively is necessary but no longer sufficient: the more useful question is *diagnostic*—why does this literature concentrate at the early stages of the translation pipeline, and what structural features (disease selection, geographic provenance, architectural choices, reporting completeness and cost transparency) predict whether a study advances toward the field? A scoping review remains the appropriate methodology, because the aim is to map and characterise a heterogeneous body of evidence rather than to estimate a pooled effect size [23, 24]; however, the synthesis is framed around an explanatory question, not only a descriptive one. The objective of this scoping review is therefore (i) to systematically map the extent and nature of research on AI-enhanced point-of-care and rapid diagnostic technologies for infectious diseases in resource-limited settings and (ii) to identify the structural features that distinguish studies that reach prospective field validation from those that remain at the laboratory or retrospective stage. Six secondary research questions address disease and modality coverage, AI techniques and deployment platforms, geographic distribution and validation levels, diagnostic accuracy, implementation barriers and evidence gaps; the explanatory layer is added through four post hoc cross-tabulations specified in Section 2.6.

2 | Methods

2.1 | Protocol and Registration

This scoping review was conducted in accordance with the JBI methodology for scoping reviews [23–25] and is reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines [26]. The review followed the five-stage JBI scoping review framework: (1) identifying the research question, (2) identifying relevant studies, (3) study selection, (4) charting the data and (5) collating, summarising and reporting the results. The protocol was registered on OSF Registries prior to screening (https://doi.org/10.17605/OSF.IO/KV8MP). No amendments to the protocol were required.

2.2 | Eligibility Criteria

Eligibility criteria were defined using the Population, Concept, Context (PCC) framework [25], as recommended by JBI for scoping reviews.

Population: Studies involving human biological specimens (blood, sputum, stool, urine and skin images), patients undergoing diagnostic testing, or health workers using AI-enhanced diagnostic tools in the context of infectious disease diagnosis.

Concept: Studies describing the development, evaluation, or deployment of AI-enhanced POC or rapid diagnostic tools for infectious diseases. Both an AI component and a POC diagnostic component were required for inclusion. The AI component was operationally defined as any use of machine learning (including deep learning, convolutional neural networks, support vector machines, random forests and vision transformers), computer vision, image classification, object detection, or algorithmic decision support that goes beyond simple threshold-based rules. The AI component must be involved in the interpretation, classification, or quality assurance of diagnostic results. The POC diagnostic component was operationally defined as tests performed at or near the site of patient care, with results available within the same clinical encounter (typically 60 min or less), including rapid diagnostic tests (RDTs), lateral flow assays (LFAs), light and smartphone-based microscopy, portable imaging (handheld ultrasound, portable x-ray and smartphone dermoscopy), molecular POC tests (e.g., GeneXpert, LAMP-based assays), biosensors and electrochemical assays. AI diagnostic systems requiring centralised laboratory infrastructure (e.g., full CT/MRI scanners, automated high-throughput laboratory analysers) that cannot be deployed at the point of care were excluded.

Context: Resource-limited settings, including LMICs as classified by the World Bank, rural or remote areas in any country, or settings explicitly described as having limited laboratory infrastructure. Studies conducted in high-income country laboratories were included only if the authors explicitly stated the tool was designed for resource-limited deployment.

Types of evidence: Peer-reviewed original research articles (all study designs including diagnostic accuracy studies, feasibility studies, randomised controlled trials, cohort studies, cross-sectional studies, qualitative studies, mixed-methods studies and implementation studies), conference proceedings and preprints were eligible. Reviews, editorials, commentaries and protocols without primary data were excluded. Studies published before January 2015, or not available in English, were excluded. The 2015 start date was selected to reflect the period of meaningful acceleration in deep learning applications to medical diagnostics following the widespread adoption of convolutional neural networks.

2.3 | Information Sources and Search Strategy

Five electronic databases were searched on 24 March 2026: PubMed/MEDLINE, Embase (via Ovid), Scopus, Web of Science (Core Collection) and IEEE Xplore. The inclusion of IEEE Xplore was a deliberate strategy to capture engineering and device-development literature not indexed in biomedical databases, which is particularly relevant for AI-enhanced diagnostic hardware. The Global Index Medicus (WHO), which aggregates regional indexes including LILACS (Latin America), AIM (Africa), IMEMR (Eastern Mediterranean), IMSEAR (South-East Asia) and WPRIM (Western Pacific), was attempted on two dates (24 March and 6 April 2026) but was inaccessible due to a prolonged server outage of the BVS search infrastructure (ECONNREFUSED on search.bvsalud.org, pesquisa.bvsalud.org and globalindexmedicus.net); this limitation is noted and discussed.

The search strategy was developed iteratively by the author and combined four concept blocks using Boolean AND: (1) artificial intelligence and machine learning terms; (2) point-of-care and rapid diagnostic terms; (3) infectious disease terms and (4) resource-limited setting terms. Each block combined controlled vocabulary (MeSH terms for PubMed, Emtree terms for Embase) with free-text synonyms searched in title, abstract and keyword fields. An initial pilot search in PubMed using a broader strategy that included terms such as 'algorithm', '*portable*', '*handheld*', '*bacter*', 'viral' and 'remote' returned 616 results with substantial noise. These broad terms were removed from all databases except Embase, which had already been executed with the original strategy (119 results; acceptable volume for screening). The refinement was documented in a change log as part of the search strategy documentation.

For PubMed, Block 1 (artificial intelligence) included the MeSH terms 'artificial intelligence', 'machine learning', 'deep learning' and 'neural networks, computer', alongside free-text terms including 'convolutional neural network', '*CNN*', '*transfer learning*', '*computer vision*', '*image recognition*', '*image classification*', '*object detection*', '*smartphone-based*' and '*vision transformer*'. Block 2 (point-of-care diagnostics) included the MeSH terms 'point-of-care testing' and 'point-of-care systems', alongside free-text terms including 'rapid diagnostic test', '*RDT*', '*lateral flow*', '*smartphone microscop*', 'mHealth', 'biosensor', '*lab-on-a-chip*' and '*microfluidic*'. Block 3 (infectious diseases) included the MeSH terms 'communicable diseases' and 'infections', alongside free-text terms covering 25 specific infectious diseases and disease categories. Block 4 (resource-limited settings) included the MeSH terms 'developing countries', 'poverty areas' and 'rural health', alongside free-text terms including 'low-resource', '*resource-limited*', '*LMIC*', 'sub-Saharan Africa', 'South Asia', 'Southeast Asia' and 'global health'. The combined search applied date limits (2015–2026) and English language restrictions.

Database-specific adaptations were made for each source. Embase searches used Emtree subject headings with explosion alongside free-text searching of title, abstract and author keywords. Scopus used TITLE-ABS-KEY field codes with truncation. Web of Science used TS (Topic) field codes, which search title, abstract, author keywords and Keywords Plus. IEEE Xplore used 'All Metadata' field codes in its Command Search interface, with simplified term lists to accommodate the platform's query length limitations. The full search strategies for all databases, including the exact syntax used and execution notes, are provided in Supporting Information S1.

2.4 | Selection of Sources of Evidence

All database exports were imported and deduplicated using a Python-based pipeline. Deduplication proceeded in two

stages: first, exact DOI matching identified and removed 510 duplicate records; second, fuzzy title matching using a similarity threshold of 0.90 identified and removed an additional 11 duplicates. This two-stage approach yielded 551 unique records from 1072 identified across the five databases. The individual database contributions were: PubMed ($n = 228$), Scopus ($n = 378$), Web of Science ($n = 283$), Embase ($n = 119$) and IEEE Xplore ($n = 64$).

Title and abstract screening was performed using a two-stage process combining automated and manual methods. First, a rule-based keyword screening algorithm was applied, classifying records against the four eligibility criteria (AI component, POC diagnostic component, infectious disease and resource-limited setting). The algorithm searched for the presence of predefined keyword sets corresponding to each criterion within the title and abstract text. Records matching all four criteria were provisionally included ($n = 247$); records matching zero or one criterion were excluded ($n = 210$); and records matching two or three criteria were flagged as uncertain ($n = 94$). Second, all 94 uncertain records were manually reviewed by the investigator, with each record classified as include or exclude based on the title and abstract. This hybrid approach was designed to maximise both efficiency and rigour: the automated component ensured consistent application of criteria across the full set, while the manual review of uncertain records ensured that nuanced judgements about eligibility were applied where automated classification was insufficiently confident.

A 20% stratified validation sample ($n = 111$; 56 from included records, 55 from excluded records) was independently re-screened to assess reliability. This validation yielded 92.8% agreement and a Cohen's kappa of 0.856, corresponding to 'almost perfect agreement' per the Landis and Koch [27] classification. The validation exceeded the protocol's pre-specified threshold of 90% consistency.

Enhanced screening was conducted on 278 records that passed title/abstract screening, combining available full texts, PubMed publication type metadata and detailed abstract review. Records were excluded at this stage if they were confirmed as reviews (identified using PubMed publication type metadata where available and title-based detection for records without metadata), were off-topic upon closer examination, or described centralised laboratory technology without a POC component. Manual review was applied to all algorithmically flagged exclusions to correct false positives, ensuring that no eligible studies were inadvertently excluded. After full-text screening, 41 records were excluded (26 reviews, 12 off-topic, 3 centralised laboratory technology), yielding 237 studies for inclusion. The complete selection process is depicted in the PRISMA-ScR flow diagram (Supporting Information S3). The full list of excluded studies with reasons is provided in Table S2.

2.5 | Data Charting

Data were extracted using a structured charting form with 19 variables aligned with the six secondary research questions: study identification (first author, year, DOI), geographic location and World Bank income classification, setting type, infectious disease, diagnostic modality, AI/ML technique, deployment platform, hardware requirements, training data characteristics, reference standard, performance metrics (sensitivity, specificity, accuracy, AUC), comparison to human interpretation, validation level, sample size, cost data, implementation barriers, funding source and commercial product status. The charting form was piloted on the first five included studies and iteratively refined, consistent with JBI guidance that scoping reviews are inherently iterative [24, 26]. The complete charting data for all 237 included studies are provided in Table S1.

Initial extraction was performed using automated natural language processing (NLP) applied to titles and abstracts. The NLP pipeline used keyword matching, pattern recognition and structured parsing to identify and extract values for each charting variable. This was enriched using full-text articles retrieved from PubMed Central (PMC) for 76 of 81 records with available PMC full texts. Full-text enrichment substantially improved completeness across key variables: AI technique extraction increased from 41% to 57%, validation level extraction from 11% to 38% and country extraction from 29% to 39%. For performance metrics, the pipeline extracted numerical values for sensitivity, specificity, accuracy and AUC using regular expression patterns applied to both abstracts and full texts.

Fields that remained incomplete after automated extraction are acknowledged as a limitation. In particular, deployment platform, cost data and implementation barriers were frequently not reported in abstracts and were only partially recoverable from full texts. The 146 studies (62%) for which PMC full texts were not available had lower extraction completeness across all variables.

Validation level was classified using a five-category schema: (1) public dataset evaluation, indicating that the AI model was trained and/or tested exclusively on publicly available benchmark datasets; (2) laboratory-only validation, indicating testing on locally acquired specimens or images under controlled laboratory conditions; (3) laboratory-based evaluation, a broader category for studies reporting evaluation in a laboratory environment without specifying the data source; (4) retrospective clinical validation, indicating that the model was tested on previously collected clinical data and (5) prospective field validation, indicating that the AI-enhanced diagnostic tool was tested in real-world conditions at or near the point of care. Studies that did not report sufficient detail to classify their validation level were categorised as 'not reported'.

2.6 | Synthesis of Results

Results were synthesised descriptively using frequency counts, cross-tabulations and visual displays. No meta-analysis was performed, consistent with scoping review methodology [28]. Diagnostic performance metrics were tabulated descriptively by disease and modality but were not pooled quantitatively due to heterogeneity in reference standards, sample sizes, evaluation protocols and validation levels. Key visualisations include a publication trend chart (Figure 1), a disease-by-modality heatmap (Figure 2), a disease-by-AI-technique

heatmap (Figure 3), geographic distribution charts colour-coded by World Bank income classification (Figure 4), a validation pipeline chart (Figure 5) and diagnostic modality distribution (Figure 6). All analyses were conducted using Python (version 3.14) with matplotlib, seaborn and pandas libraries. The analysis scripts are available in the project repository. The completed PRISMA-ScR checklist is provided in Supporting Information S2.

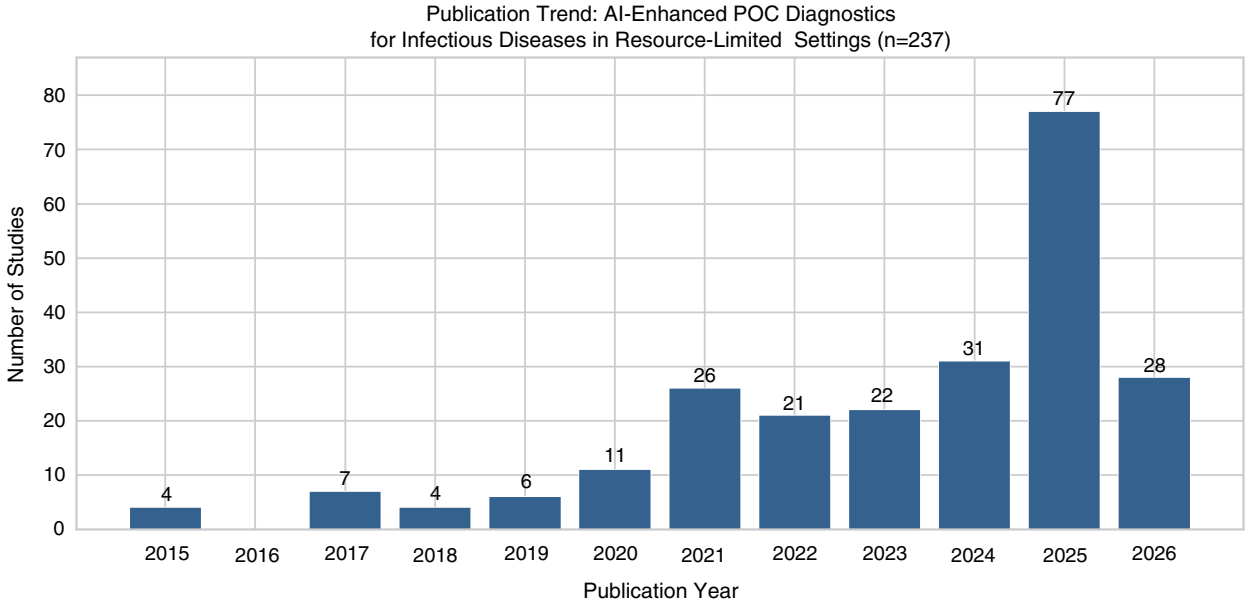


FIGURE 1 | Publication trend of AI-enhanced point-of-care diagnostic studies for infectious diseases in resource-limited settings, 2015–2026 ($n = 237$). The 2025–2026 period accounts for 44% of all included studies.

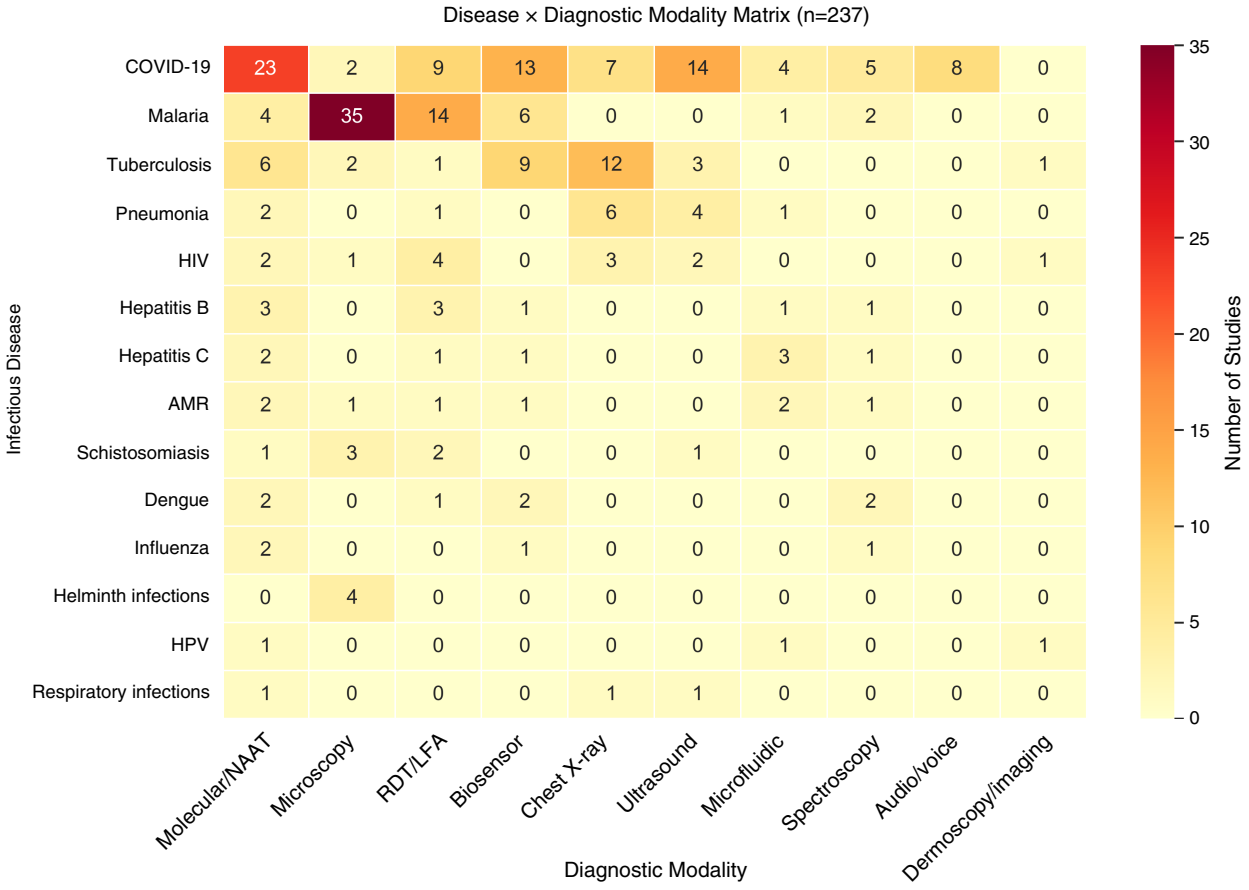


FIGURE 2 | Disease-by-diagnostic modality heatmap showing the number of studies at each intersection ($n = 237$). Malaria/microscopy ($n = 35$) and COVID-19/molecular diagnostics ($n = 23$) are the most densely studied combinations.

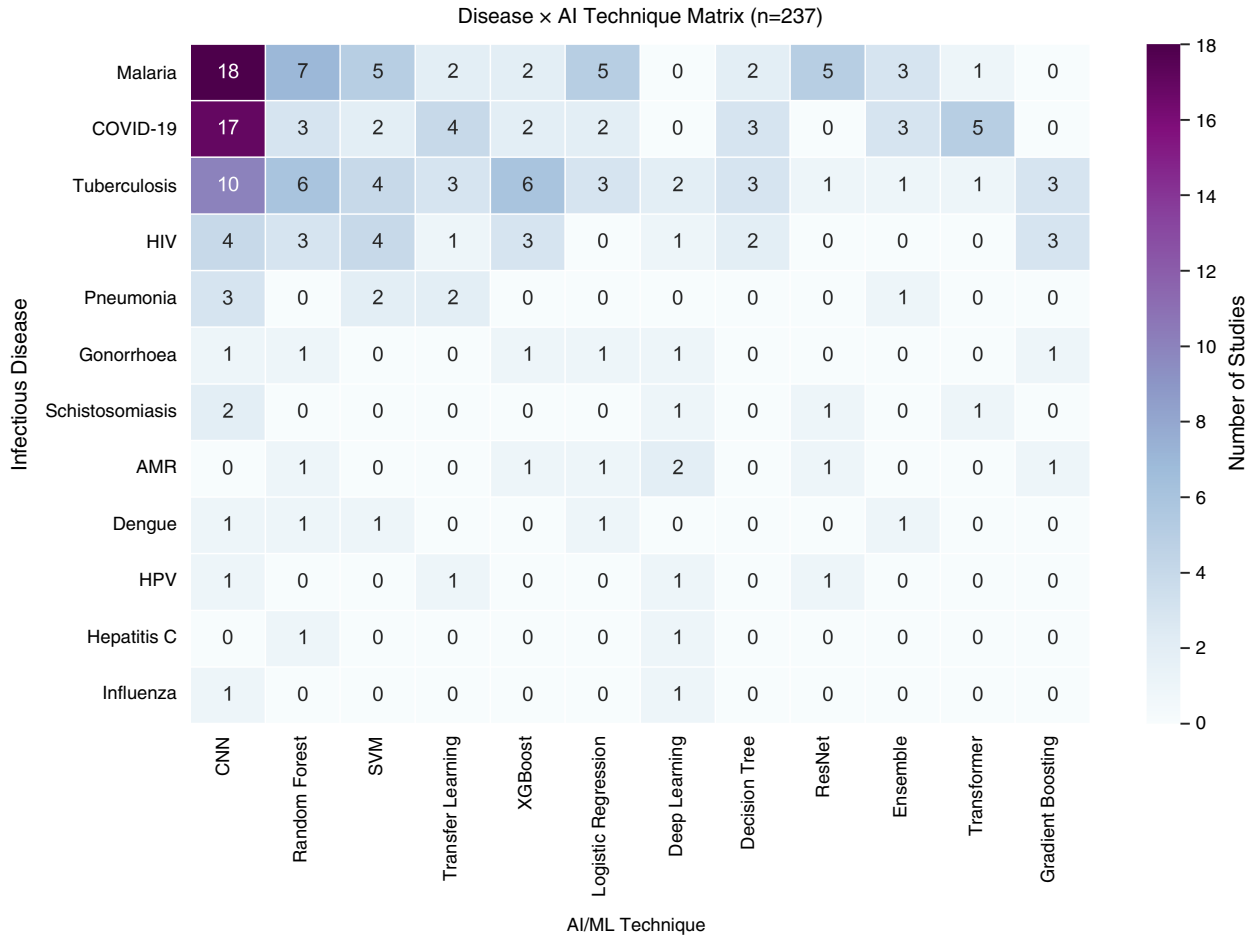


FIGURE 3 | Disease-by-AI-technique heatmap showing the number of studies at each intersection ($n = 237$). CNN usage is concentrated in malaria ($n = 18$) and COVID-19 ($n = 17$) studies.

To address the explanatory question stated in Section 1, four pre-specified post hoc cross-tabulations were generated on the final 237-study extraction set without re-screening: (i) disease × validation level and World Bank income class × validation level (Figures 7 and 8); (ii) reporting-completeness sensitivity—comparison of disease and AI-technique distributions in the complete-case subset (studies reporting country, deployment platform and validation level) versus the missing-case subset (Figures 9–11); (iii) architecture weight class (heavy/medium/lightweight/generic/non-DL) × validation level for the top 50 studies by OpenAlex citation count, with a full-corpus robustness check (Figure 12) and (iv) cost-data reporting × validation level (Figure 13). These analyses were not pre-registered in the original protocol; they are reported as post hoc explanatory cross-tabulations and are constrained by the same charting completeness as the descriptive analyses.

3 | Results

3.1 | Search Results

The database searches identified 1072 records: PubMed ($n = 228$), Scopus ($n = 378$), Web of Science ($n = 283$), Embase ($n = 119$) and IEEE Xplore ($n = 64$). The Global Index Medicus was inaccessible due to a prolonged server outage and could not be searched. After automated deduplication (521 duplicates removed by DOI and fuzzy title matching), 551 unique records entered title and abstract screening. Of these, 278 were included after title/abstract screening (273 excluded), and 237 were included after full-text screening (41 excluded: 26 reviews, 12 off-topic, 3 centralised laboratory technology). The PRISMA-ScR flow diagram is presented in Supporting Information S3.

3.2 | Publication Trends

The 237 included studies span 2015 to early 2026, with marked exponential growth (Figure 1). The pre-pandemic period (2015–2019) contributed only 21 studies (8.9%). Publication volume surged during and after the COVID-19 pandemic, with 58 studies (24.5%) published in 2020–2022 and 53 studies (22.4%) in 2023–2024. The most recent period (2025–2026) accounts for 105 studies (44.3%), indicating that nearly half of all research in this field has been published within the last 18 months. This trajectory reflects the convergence of pandemic-driven diagnostic urgency, maturing deep learning frameworks and the proliferation of low-cost smartphone and edge computing hardware.

The growth was not uniform across diseases or modalities. COVID-19-related publications accounted for a disproportionate

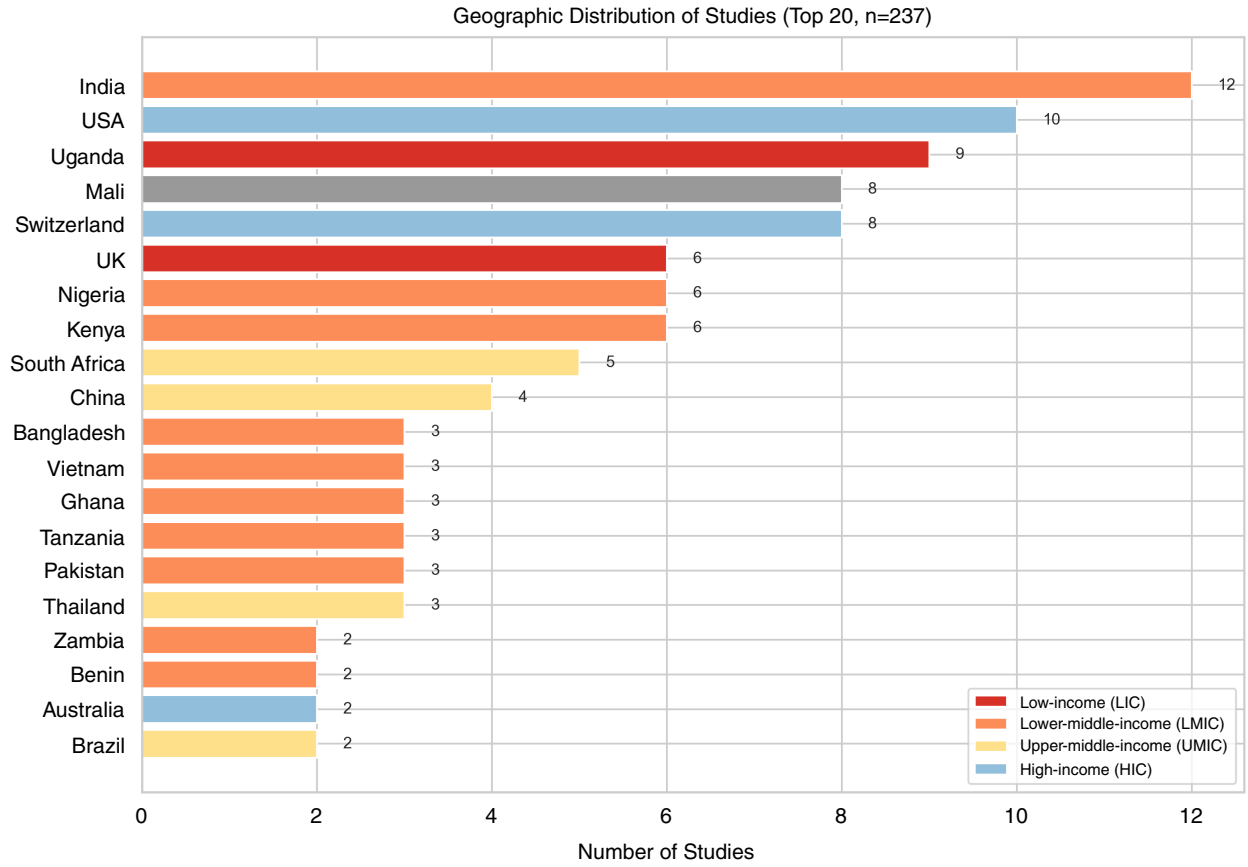


FIGURE 4 | Geographic distribution of studies by country (top 20), colour-coded by World Bank income classification ($n = 92$ studies with extractable country data). Red = low-income country (LIC), orange = lower-middle-income country (LMIC), yellow = upper-middle-income country (UMIC), and blue = high-income country (HIC).

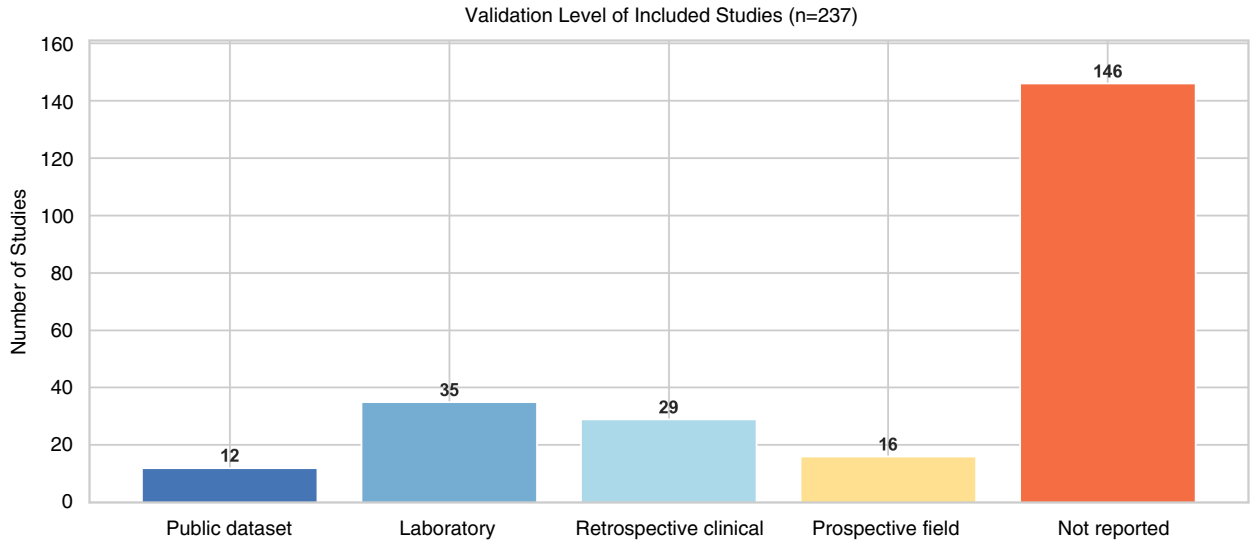


FIGURE 5 | Validation level of included studies ($n = 237$). Only 16 studies (6.8%) report prospective field validation; 146 studies (62%) did not report their validation level.

share of the 2020–2022 surge, while malaria and tuberculosis studies have shown steadier growth. The 2025–2026 period shows a diversification trend, with increasing representation of biosensor-based diagnostics, antimicrobial resistance (AMR) testing and neglected tropical diseases, though these remain a minority of the literature.

3.3 | Disease Coverage

COVID-19 was the most frequently studied disease ($n = 76$, 32.1%), followed by malaria ($n = 64$, 27.0%), tuberculosis ($n = 33$, 13.9%), HIV ($n = 19$, 8.0%) and pneumonia ($n = 15$, 6.3%) (Table 1). Together, these five conditions accounted for 87%

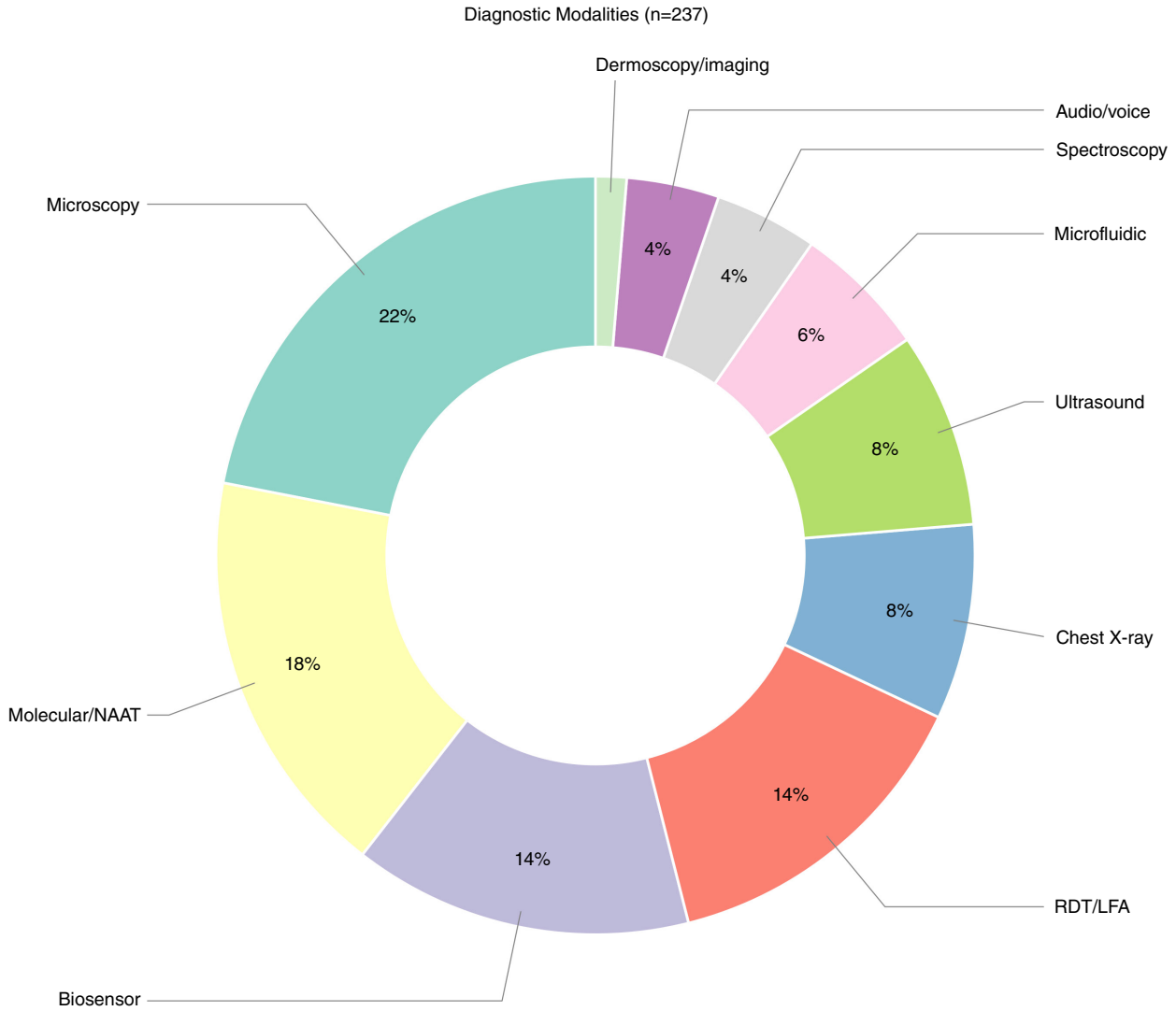


FIGURE 6 | Diagnostic modality distribution across included studies ($n = 237$). Microscopy (22%), molecular/NAAT (18%), biosensors (14%) and RDTs/lateral flow assays (14%) are the four most common modalities.

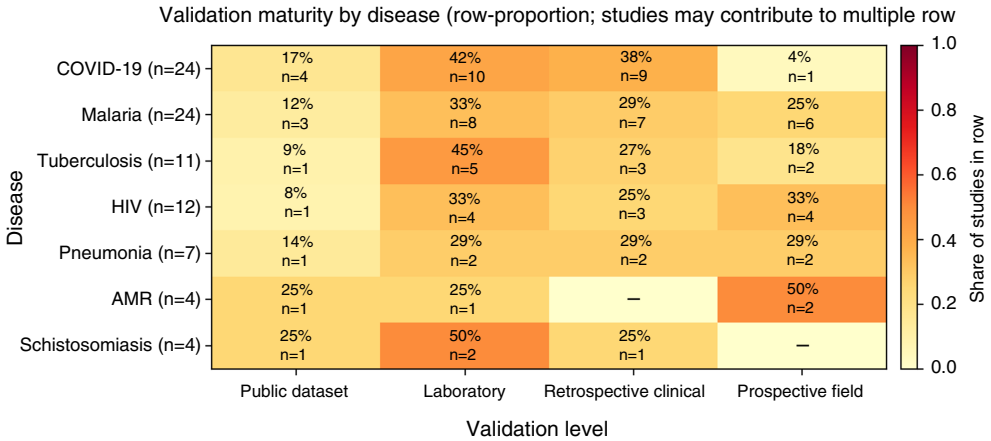


FIGURE 7 | Validation maturity by disease (row-proportion heatmap). Restricted to the seven diseases represented by at least four studies with extractable validation data ($n = 83$ studies; multi-coded studies contribute to each disease row in which they appear). Cell values show the proportion of studies in each row reaching the given validation level, with the underlying count beneath; row totals appear in parentheses next to the disease label. COVID-19 ($n = 24$ studies with validation data) clusters at the laboratory and retrospective stages; malaria ($n = 24$) distributes more evenly; HIV ($n = 12$) reaches prospective field validation at the highest rate among well-represented diseases; tuberculosis ($n = 11$) remains predominantly at the laboratory and retrospective stages.

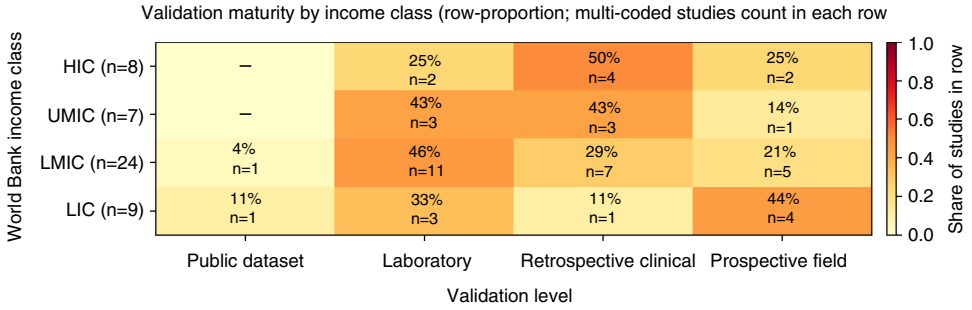


FIGURE 8 | Validation maturity by World Bank income class (row-proportion heatmap). Restricted to studies reporting both income class and validation level ($n = 48$). HIC-affiliated studies ($n = 8$) concentrate at later validation stages (75% retrospective or prospective); LIC-affiliated studies ($n = 9$) report the highest prospective-field share (44%); LMIC ($n = 24$) and UMIC ($n = 7$) sit between. Sample sizes per stratum are small (corpus has 71% missingness on income class) and the pattern is reported as a structural observation rather than a statistical claim. The data are inconsistent with the conventional 'HIC develops, LMIC tests' framing.

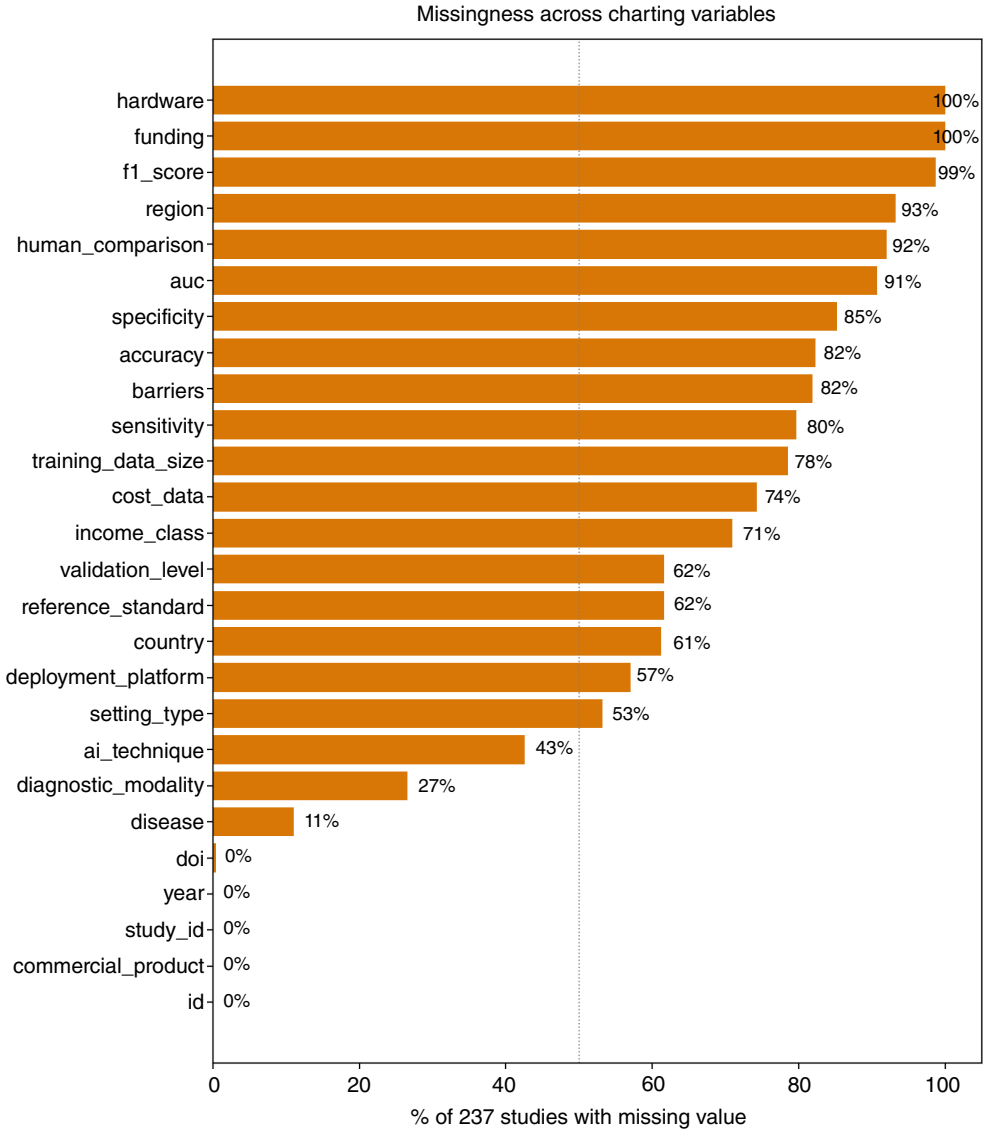


FIGURE 9 | Charting-variable missingness across the 237-study corpus. Horizontal bar chart of the percentage of studies missing each charting variable, sorted from most-missing to least-missing. The three core operational variables flagged by peer review—country (61% missing), deployment platform (57% missing) and validation level (62% missing)—appear as a cluster in the mid-range. The reference vertical line is at 50% missingness.

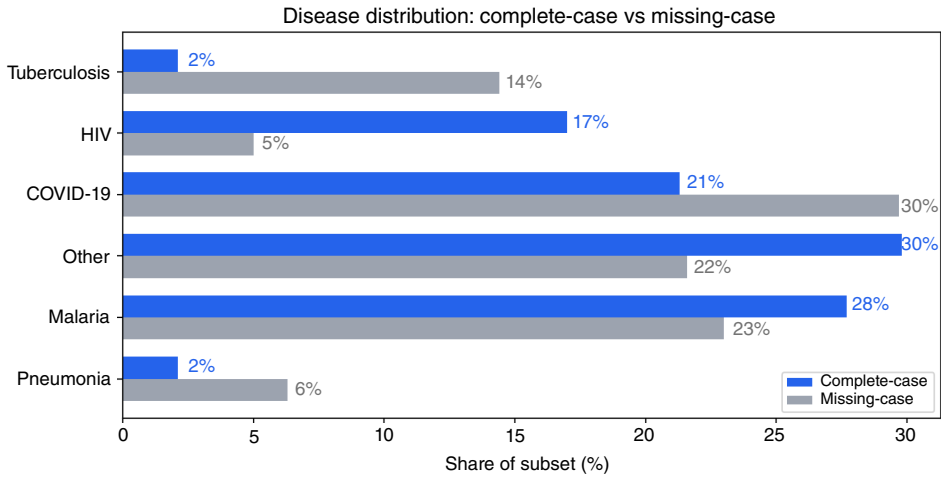


FIGURE 10 | Disease distribution: Complete-case versus missing-case subsets. Paired horizontal bar chart comparing disease distribution in the complete-case subset ($n = 40$ studies reporting country + deployment platform + validation level) against the missing-case subset ($n = 197$ studies missing at least one of these three variables). HIV is over-represented in the complete-case subset by 12 percentage points (17% vs. 5%); tuberculosis is under-represented by 12 percentage points (2% vs. 14%); COVID-19 is under-represented by 8 percentage points (21% vs. 30%).

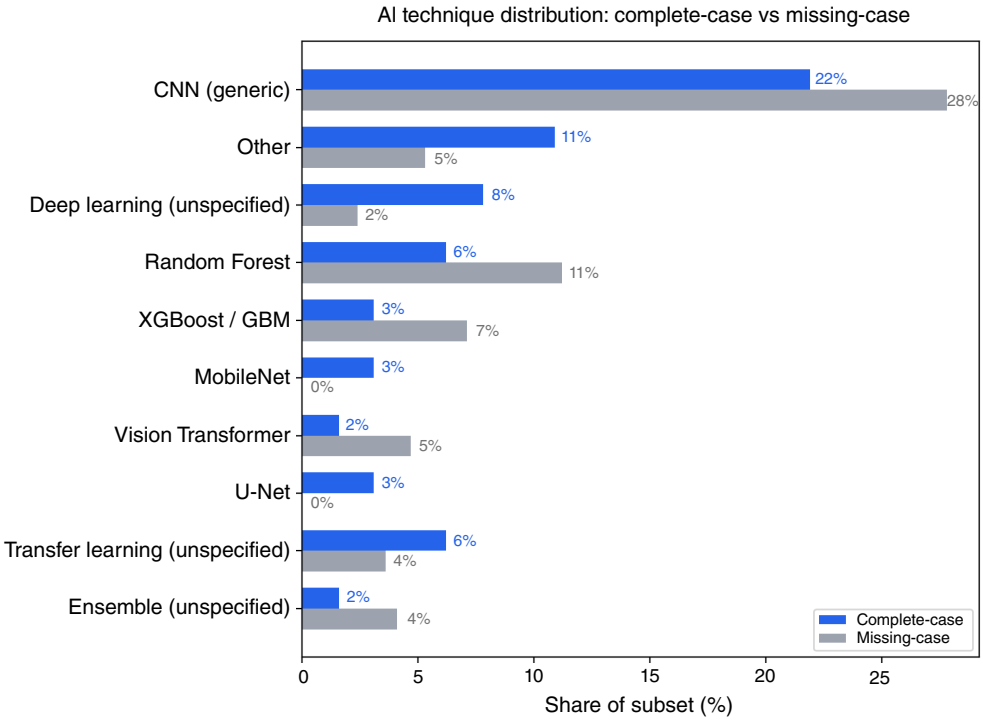


FIGURE 11 | AI-technique distribution: Complete-case versus missing-case subsets. Paired horizontal bar chart comparing the AI/ML technique distribution in the complete-case subset ($n = 40$) against the missing-case subset ($n = 197$). Maximum absolute between-subset difference is 6 percentage points (Generic CNN/DL: 22% vs. 28%); the Section 3.5 picture of technique dominance is therefore robust to reporting selection, in contrast to the disease coverage figures (Figure 10).

of the disease-specific literature. The remaining 13% was distributed across a long tail of diseases: antimicrobial resistance ($n = 8$, 3.4%), schistosomiasis ($n = 7$, 3.0%), dengue ($n = 7$, 3.0%), influenza ($n = 5$, 2.1%), hepatitis C ($n = 5$, 2.1%), respiratory infections ($n = 5$, 2.1%), hepatitis B ($n = 4$, 1.7%), helminth infections ($n = 4$, 1.7%), HPV ($n = 3$, 1.3%), leishmaniasis ($n = 3$, 1.3%), gonorrhoea ($n = 2$, 0.8%), cholera ($n = 2$, 0.8%), Ebola ($n = 2$, 0.8%) and typhoid ($n = 2$, 0.8%).

Neglected tropical diseases were represented by schistosomiasis ($n = 7$), dengue ($n = 7$), leishmaniasis ($n = 3$), helminth infections ($n = 4$) and opisthorchiasis ($n = 1$), collectively accounting for fewer than 8% of studies. Sexually transmitted infections other than HIV—including gonorrhoea ($n = 2$), HPV ($n = 3$) and syphilis—were under-represented, as were fungal infections. Chagas disease, Zika virus and trachoma are absent from the literature entirely, despite all three appearing

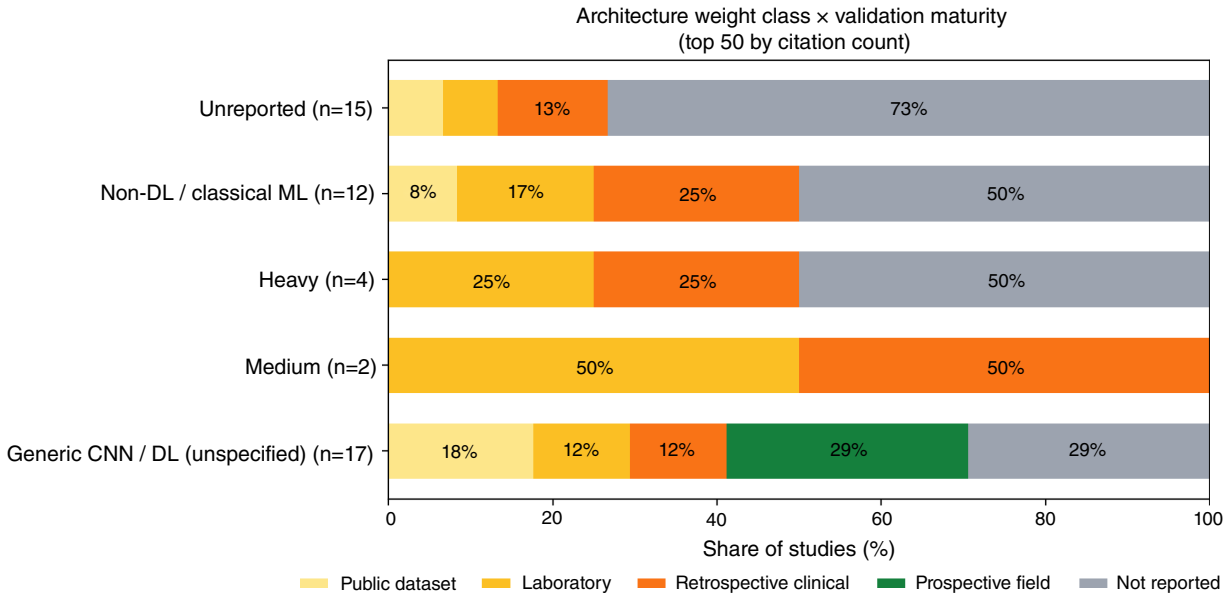


FIGURE 12 | Architecture weight class × validation maturity, top 50 by citation count. Stacked horizontal bar chart showing the proportion of top-50-cited studies reaching each validation level, stratified by architecture weight class. The 50 most-cited studies in the corpus contain no explicitly lightweight architectures (MobileNet, YOLO-tiny, SqueezeNet, EfficientNet-Lite, MobileViT, ShuffleNet); the top-50 composition is 17 generic CNN/deep-learning studies (untyped), 15 architecture-unreported studies, 12 non-deep-learning/classical-ML studies, 4 heavy-architecture studies (ResNet, EfficientNet B-series, Vision Transformer family) and 2 medium-architecture studies (DenseNet, U-Net). Among the top 50, only the generic CNN studies reach prospective field validation (5 of 17, 29%); the 4 heavy-architecture and 2 medium-architecture top-50 papers all remain at laboratory, retrospective clinical, or unreported validation stages. In the full-corpus robustness check (235 studies with citation data, not shown), lightweight architectures ($n = 3$ in the corpus) reach prospective field validation in 33% of cases (1 of 3), while heavy architectures ($n = 21$) reach 0%.

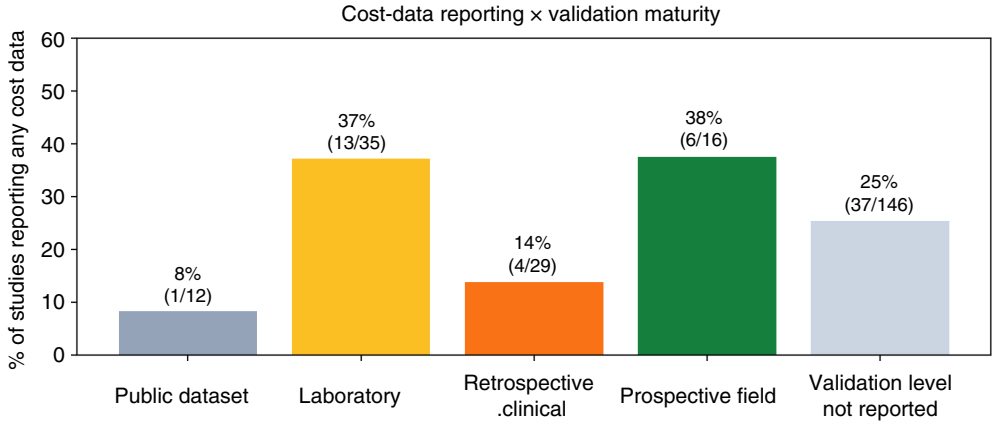


FIGURE 13 | Cost-data reporting × validation maturity. Bar chart of the percentage of studies reporting any cost data (per-test cost, hardware cost or comparator cost), stratified by validation level. Counts beneath each bar show studies with cost data over total studies at that validation level. Cost reporting is essentially flat across the validation pipeline: 8% (1/12) at the public dataset stage, 37% (13/35) at laboratory, 14% (4/29) at retrospective clinical, 38% (6/16) at prospective field and 25% (37/146) among studies with unreported validation level. Cost reporting does not mature with the validation pipeline.

on the WHO neglected tropical diseases list with substantial LMIC burden.

The disease-by-modality matrix (Figure 2) reveals distinct technology-disease pairings that reflect established clinical workflows. Malaria research is concentrated in microscopy ($n = 35$), reflecting the central role of blood smear analysis in malaria diagnosis and the natural fit of AI-powered image recognition for parasite detection and quantification. COVID-19 spans multiple modalities including molecular diagnostics ($n = 23$), ultrasound ($n = 14$), biosensors ($n = 13$) and chest x-ray ($n = 9$), reflecting the urgency-driven breadth of diagnostic innovation during the pandemic. Tuberculosis research centres on chest x-ray ($n = 12$) and molecular diagnostics ($n = 9$), aligning with the WHO-endorsed AI-assisted computer-aided detection (CAD) pathway for TB screening. HIV studies are distributed across RDTs/lateral flow assays ($n = 3$), molecular tests ($n = 2$) and multiple other modalities including chest x-ray and dermoscopy/imaging, reflecting the diversity of HIV diagnostic and monitoring needs. Schistosomiasis research spans microscopy,

TABLE 1 | Disease distribution of included studies ($N = 237$).

Disease	*n* (%)	Diagnostic modalities	AI/ML techniques
COVID-19	76 (32.1%)	Audio/voice; Biosensor; Chest x-ray; Microfluidic; Microscopy; Molecular/NAAT; RDT/LFA; Spectroscopy; Ultrasound	ANN; ANN/MLP; Attention Mechanism; CNN; Decision Tree; DenseNet; EfficientNet; Ensemble; LSTM; Logistic Regression; Mask R-CNN; Naive Bayes; Random Forest; SVM; Transfer Learning; Transformer; U-Net; VGG; Vision Transformer; XGBoost; k-NN
Malaria	64 (27.0%)	Biosensor; Microfluidic; Microscopy; Molecular/NAAT; RDT/LFA; Spectroscopy	ANN/MLP; CNN; Decision Tree; DenseNet; Elastic Net; Ensemble; Faster R-CNN; Federated Learning; GoogLeNet; Inception; LASSO; LSTM/RNN; Logistic Regression; Mask R-CNN; MobileNet; Random Forest; ResNet; Ridge Regression; SVM; Transfer Learning; Transformer; VGG; Vision Transformer; XGBoost; YOLO
Tuberculosis	33 (13.9%)	Biosensor; Chest x-ray; Dermoscopy/imaging; Microscopy; Molecular/NAAT; RDT/LFA; Ultrasound	CNN; Decision Tree; Deep Learning; EfficientNet; Ensemble; Gradient Boosting; Inception; LASSO; Logistic Regression; Naive Bayes; RNN; Random Forest; ResNet; SVM; Transfer Learning; Transformer; XGBoost
HIV	19 (8.0%)	Chest x-ray; Dermoscopy/imaging; Microscopy; Molecular/NAAT; RDT/LFA; Ultrasound	CNN; Decision Tree; Deep Learning; Gradient Boosting; LASSO; MobileNet; Random Forest; SVM; Transfer Learning; XGBoost
Pneumonia	15 (6.3%)	Chest x-ray; Microfluidic; Molecular/NAAT; RDT/LFA; Ultrasound	ANN/MLP; CNN; EfficientNet; Ensemble; SVM; Transfer Learning
AMR	8 (3.4%)	Biosensor; Microfluidic; Microscopy; Molecular/NAAT; RDT/LFA; Spectroscopy	Deep Learning; Gradient Boosting; Logistic Regression; Random Forest; ResNet; XGBoost
Schistosomiasis	7 (3.0%)	Microscopy; Molecular/NAAT; RDT/LFA; Ultrasound	CNN; Deep Learning; Feature Pyramid Network; Mask R-CNN; ResNet; Transformer; U-Net
Dengue	7 (3.0%)	Biosensor; Molecular/NAAT; RDT/LFA; Spectroscopy	CNN; Ensemble; Logistic Regression; Random Forest; SVM; U-Net
Influenza	5 (2.1%)	Biosensor; Molecular/NAAT; Spectroscopy	CNN; Deep Learning; RNN
Hepatitis C	5 (2.1%)	Biosensor; Microfluidic; Molecular/NAAT; RDT/LFA; Spectroscopy	ANN/MLP; Deep Learning; Random Forest
Respiratory infections	5 (2.1%)	Chest x-ray; Molecular/NAAT; Ultrasound	CNN; EfficientNet; Transfer Learning
Hepatitis B	4 (1.7%)	Biosensor; Microfluidic; Molecular/NAAT; RDT/LFA; Spectroscopy	Deep Learning
Helminth infections	4 (1.7%)	Microscopy	CNN; Deep Learning; Transfer Learning
HPV	3 (1.3%)	Dermoscopy/imaging; Microfluidic; Molecular/NAAT	CNN; Deep Learning; ResNet; Transfer Learning
Leishmaniasis	3 (1.3%)	Biosensor; Microscopy	Self-Organising Map
Gonorrhoea	2 (0.8%)	Molecular/NAAT; RDT/LFA	CNN; Deep Learning; Gradient Boosting; Logistic Regression; MobileNet; Random Forest; XGBoost; YOLO

(Continues)

TABLE 1 | (Continued)

Disease	*n* (%)	Diagnostic modalities	AI/ML techniques
Cholera	2 (0.8%)	Biosensor; Molecular/NAAT	
Ebola	2 (0.8%)		Logistic Regression
Typhoid	2 (0.8%)		ANN/MLP; SVM

Note: Modalities and techniques are listed for each disease category; individual studies may report multiple modalities or techniques.
Abbreviations: AMR, antimicrobial resistance; ANN, artificial neural network; CNN, convolutional neural network; HPV, human papillomavirus; LAMP, loop-mediated isothermal amplification; LFA, lateral flow assay; LSTM, long short-term memory; MLP, multilayer perceptron; NAAT, nucleic acid amplification test; RDT, rapid diagnostic test; RNN, recurrent neural network; SVM, support vector machine; YOLO, You Only Look Once.

molecular/NAAT, RDT/LFA and ultrasound modalities; AMR studies use biosensors, microfluidics, microscopy, molecular diagnostics and spectroscopy.

3.4 | Diagnostic Modalities

The 10 identified diagnostic modalities exhibit a long-tail distribution (Figure 6). Microscopy was the most common ($n = 50$, 21.1%), driven predominantly by malaria blood smear analysis but also including applications in helminth detection, schistosomiasis egg identification and tuberculosis sputum microscopy. Molecular diagnostics and nucleic acid amplification tests ranked second ($n = 40$, 16.9%), encompassing CRISPR-based assays, loop-mediated isothermal amplification (LAMP), GeneXpert adaptations and RT-PCR-based systems enhanced with AI interpretation. Notable examples include the study by Xu et al. [29], which combined molecular/NAAT with microfluidic technology for HPV detection, achieving 94.0% accuracy on a sample of 10,976 specimens using a ResNet architecture and the work of Yang et al. [30], which applied CNN, random forest and SVM algorithms to molecular TB diagnostics, reporting 97.0% sensitivity and 98.1% specificity.

Biosensors ($n = 33$, 13.9%) represented a rapidly growing modality, particularly for COVID-19, malaria and AMR detection. These include electrochemical, optical and surface plasmon resonance-based sensors enhanced with AI for signal interpretation and noise reduction. RDTs/lateral flow assays ($n = 32$, 13.5%) were closely matched, with AI typically applied to the automated interpretation of test lines via smartphone camera image analysis. For example, Turbé et al. [10] developed a CNN and SVM-based system for automated reading of HIV rapid diagnostic tests, achieving 95.9% sensitivity and 99.0% specificity in a prospective field validation involving 11,374 tests—one of the largest field validation studies in this review. Similarly, Bahati et al. [31] reported 96.1% sensitivity and 98.0% specificity for AI-assisted malaria RDT interpretation, while Lindblade et al. [32] achieved 96.8% accuracy in a prospective field study of malaria RDT reading using logistic regression.

Chest x-ray and ultrasound each contributed 19 studies (8.0%), reflecting the growing role of portable imaging in AI-enhanced diagnostics. Chest x-ray applications were dominated by TB screening studies, including the work of Weber et al. [33], who combined chest x-ray with ultrasound and molecular diagnostics using random forest algorithms for TB and HIV co-infection screening, reporting 95.0% sensitivity and 95.0% specificity in a prospective field validation of 541 patients. Kessler et al. [34] demonstrated AI-enhanced point-of-care ultrasound for pneumonia detection with 88% sensitivity and 89% specificity in a study of 604 patients involving both prospective field and retrospective clinical components.

Emerging modalities included microfluidics ($n = 13$, 5.5%), which integrate sample preparation, amplification and detection on a single chip with AI-driven result interpretation; spectroscopy ($n = 10$, 4.2%), including Raman and infrared spectroscopy coupled with machine learning for pathogen identification; audio/voice-based screening ($n = 9$, 3.8%), a novel modality that uses AI analysis of cough sounds, breathing patterns, or voice characteristics to screen for respiratory infections including COVID-19 and tuberculosis; and dermoscopy/colposcopy ($n = 3$, 1.3%), applied to HPV-related cervical lesions and HIV-associated skin conditions.

3.5 | AI and Machine Learning Techniques

Convolutional neural networks (CNNs) were the dominant AI technique, reported in 61 studies (25.7%), consistent with the predominance of image-based diagnostic tasks (microscopy, radiology, RDT image interpretation) (Figure 14; Table 2). Classical machine learning algorithms remain widely used: random forests ($n = 23$, 9.7%), support vector machines ($n = 18$, 7.6%), XGBoost ($n = 13$, 5.5%), logistic regression ($n = 12$, 5.1%) and decision trees ($n = 10$, 4.2%). The persistence of classical algorithms alongside deep learning reflects the practical reality that simpler models are often more suitable for resource-constrained deployment: they require less computational power, are more interpretable and can run on low-specification hardware.

Specific deep learning architectures reported include ResNet ($n = 10$, 4.2%), transfer learning approaches ($n = 10$, 4.2%), YOLO for object detection ($n = 5$), DenseNet ($n = 3$), EfficientNet ($n = 3$), MobileNet ($n = 3$), Inception ($n = 3$), VGG ($n = 3$) and vision transformers ($n = 3$). Ensemble methods combining multiple algorithms were reported in 8 studies (3.4%). Recurrent neural networks (RNNs and LSTMs) were used in temporal data analysis applications, including cough audio classification and biosensor signal processing.

The disease-by-AI-technique matrix (Figure 3) reveals important patterns. CNN usage is concentrated in malaria ($n = 18$) and COVID-19 ($n = 17$) studies, consistent with the dominance of image-based tasks in these disease areas. TB studies employ a more diverse range of techniques including random forests ($n = 6$), SVMs ($n = 4$) and XGBoost ($n = 3$), reflecting the variety of diagnostic modalities and data types involved in TB diagnosis. HIV studies similarly show technique diversity, with CNNs, decision trees, gradient boosting, LASSO, MobileNet, random forests, SVMs,

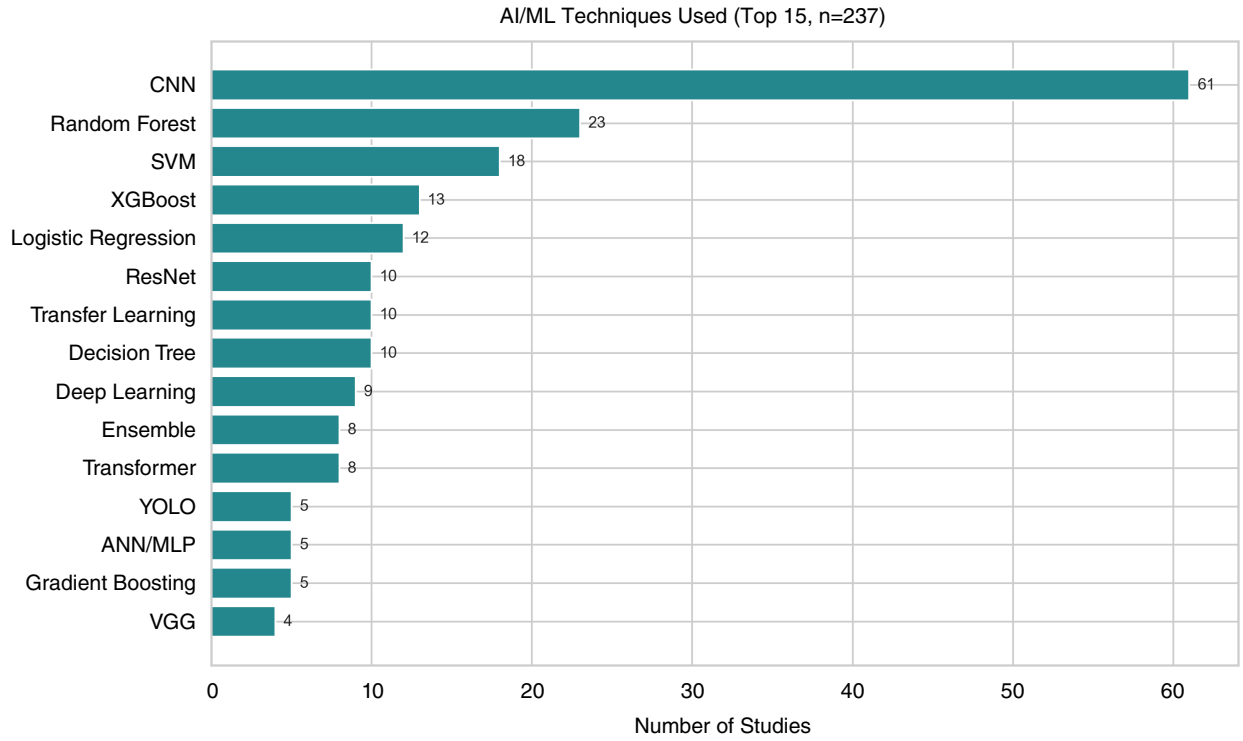


FIGURE 14 | AI and machine learning techniques used across included studies, ranked by frequency (top 15, $n = 237$). Convolutional neural networks (CNNs) are the dominant technique ($n = 61$), followed by random forests ($n = 23$) and support vector machines ($n = 18$).

transfer learning and XGBoost all represented across 19 studies. Malaria research demonstrates the broadest AI technique diversity, with 24 distinct algorithms or architectures reported across 64 studies, including emerging approaches such as federated learning, vision transformers and YOLO-based object detection.

Transfer learning was explicitly reported in 10 studies, predominantly using ImageNet-pretrained models fine-tuned on medical imaging data. This approach is particularly relevant for resource-limited settings where labelled training data may be scarce. Notable examples include Holmström et al. [35], who applied GoogLeNet with transfer learning to malaria microscopy, achieving 89% sensitivity and von Bahr et al. [20], who used CNN with transfer learning for helminth detection, reporting 61.1% sensitivity and 98.7% specificity on 777 specimens. The relatively low sensitivity in the latter study highlights the challenge of applying transfer learning across large domain gaps between natural images and medical specimens.

Notably, newer architectures such as vision transformers ($n = 3$), federated learning frameworks ($n = 1$) and self-supervised learning approaches remain rare, each appearing in fewer than five studies. Federated learning, which enables model training across distributed sites without centralising sensitive patient data, appeared in only one malaria study despite its clear relevance to multi-site LMIC deployments. Generative AI and large language model-based approaches were absent from the included literature.

3.6 | Geographic Distribution

A note on terminology: throughout this review, 'LMIC' is used in the broader development-economics sense (LIC + LMIC + UMIC) except where explicitly classified against World Bank income groups, in which case 'LMIC' refers to the World Bank's lower-middle-income category specifically. The Section 3.12.1 post hoc income-class cross-tabulation reports HIC, UMIC, LMIC and LIC as separate strata.

Country-level data were extractable for 92 of 237 studies (39%), with the remainder not reporting sufficient geographic detail in titles, abstracts, or available full texts for classification. Among the 92 studies with extractable data, India ($n = 12$), the United States ($n = 10$) and Uganda ($n = 9$) were the most frequently represented countries (Figure 4). Mali ($n = 8$), Switzerland ($n = 8$) and the United Kingdom ($n = 6$) also featured prominently, reflecting both LMIC field sites and HIC-based technology development laboratories. Nigeria ($n = 6$) and Kenya ($n = 6$) were the most frequently represented sub-Saharan African countries after Uganda.

East African countries collectively contributed the largest share of LMIC-based studies, including Uganda ($n = 9$), Kenya ($n = 6$), Tanzania ($n = 3$) and Rwanda ($n = 1$). West Africa was represented by Nigeria ($n = 6$), Mali ($n = 8$), Ghana ($n = 3$), Senegal ($n = 2$), Benin ($n = 2$), Burkina Faso ($n = 1$), Gambia ($n = 1$), Liberia ($n = 1$) and Sierra Leone ($n = 1$). South Asian contributions came primarily from India ($n = 12$), Bangladesh ($n = 3$) and Pakistan ($n = 3$). Southern Africa was represented by South Africa ($n = 5$) and Zambia ($n = 2$). Southeast Asian representation included Vietnam ($n = 3$), Thailand ($n = 3$), Indonesia ($n = 1$) and the Philippines ($n = 1$). Latin America was represented only by Brazil ($n = 2$) and Peru ($n = 1$).

When classified by World Bank income group, lower-middle-income countries (LMIC, $n = 36$) contributed the most studies,

TABLE 2 | Summary characteristics of included studies ($N = 237$).

Characteristic	Category	*n*	%
Publication period	2015–2019	21	8.9
	2020–2022	58	24.5
	2023–2024	53	22.4
	2025–2026	105	44.3
Disease	COVID-19	76	32.1
	Malaria	64	27.0
	Tuberculosis	33	13.9
	HIV	19	8.0
	Pneumonia	15	6.3
	AMR	8	3.4
	Schistosomiasis	7	3.0
	Dengue	7	3.0
	Influenza	5	2.1
	Hepatitis C	5	2.1
Diagnostic modality	Microscopy	50	21.1
	Molecular/NAAT	40	16.9
	Biosensor	33	13.9
	RDT/LFA	32	13.5
	Chest x-ray	19	8.0
	Ultrasound	19	8.0
	Microfluidic	13	5.5
	Spectroscopy	10	4.2
	Audio/voice	9	3.8
	Dermoscopy/imaging	3	1.3
AI technique	CNN	61	25.7
	Random Forest	23	9.7
	SVM	18	7.6
	XGBoost	13	5.5
	Logistic Regression	12	5.1
	ResNet	10	4.2
	Transfer Learning	10	4.2
	Decision Tree	10	4.2
	Deep Learning	9	3.8
	Ensemble	8	3.4

Note: Top 10 categories shown for disease, diagnostic modality and AI technique. Individual studies may contribute to multiple categories.
Abbreviations: CNN, convolutional neural network; LFA, lateral flow assay; NAAT, nucleic acid amplification test; RDT, rapid diagnostic test; SVM, support vector machine.

followed by high-income countries (HIC, $n = 18$), upper-middle-income countries (UMIC, $n = 12$) and low-income countries (LIC, $n = 12$) (Figures 4 and 15). The presence of HIC-based studies reflects laboratories developing tools intended for LMIC deployment—for example, Swiss-based studies developing malaria microscopy algorithms for field use in West Africa and US-based studies developing molecular diagnostic platforms intended for resource-limited deployment. This motivates the validation level analysis (Section 3.8), as HIC-developed tools require rigorous field validation in the target deployment settings.

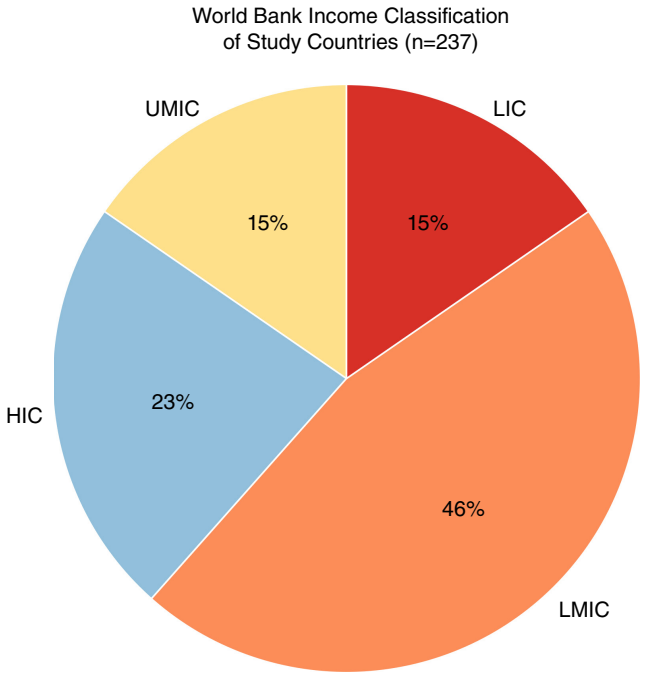


FIGURE 15 | World Bank income classification of study countries ($n = 92$ studies with extractable data). Lower-middle-income countries contribute the largest share.

Geographic gaps are notable. West African countries other than Nigeria, Mali and Ghana are significantly underrepresented despite bearing a disproportionate burden of malaria and neglected tropical diseases. Central Africa (Democratic Republic of Congo, Cameroon, Central African Republic) is almost entirely absent from the literature. Central Asia, the Pacific Islands and the Caribbean are not represented in any included study. Within Latin America, only Brazil and Peru appear, despite the region's burden of dengue, Chagas disease, leishmaniasis and other tropical infections. The Ebola-affected countries of West Africa (Liberia, Sierra Leone) each appear in only one study, specifically in the context of Ebola diagnostic research.

3.7 | Deployment Platforms

Among the 102 studies (43%) reporting deployment information, smartphone-based applications were the most common platform, reflecting the ubiquity and computational capabilities of modern smartphones. Smartphone-based systems leverage the device's camera for image capture (microscopy, RDT reading, dermoscopy), its processor for on-device model inference, and its connectivity for cloud-based analysis when available. Web/cloud-based systems were the second most common platform, offering greater computational power but requiring internet connectivity—a significant constraint in many LMIC

settings. Embedded and edge devices constituted a smaller but notable category, including implementations on Raspberry Pi single-board computers and FPGA (field-programmable gate array) hardware, which offer low-cost, low-power alternatives for running AI models without smartphone or internet dependency.

The remaining 135 studies (57%) did not explicitly report their deployment platform (Figure 16), suggesting that many remain at the algorithm development stage without a defined path to clinical implementation. This finding is consistent with the validation level analysis (Section 3.8) and indicates that the majority of studies focus on demonstrating algorithmic feasibility rather than addressing the engineering and implementation challenges of real-world deployment.

Several studies reported notable deployment platform innovations. Guo et al. [36] described a microfluidic-based malaria diagnostic with CNN-powered analysis running on a smartphone, achieving 97.83% accuracy in a prospective field validation. Prathap and Yadav [37] deployed a YOLO-based malaria microscopy system achieving 99.75% accuracy in a prospective field setting. Afifi et al. [38] reported a CNN-based ensemble system for COVID-19 and pneumonia detection from chest x-rays, achieving 91.2% accuracy on 1311 images in a prospective field study.

3.8 | Validation Levels

The validation level analysis reveals a striking gap (Figure 5). Among the 91 studies (38%) with extractable validation data, retrospective clinical validation was the most common ($n = 29$, 12.2% of all studies), followed by laboratory-only validation ($n = 23$, 9.7%), prospective field validation ($n = 16$, 6.8%), laboratory-based evaluation ($n = 12$, 5.1%) and public dataset evaluation ($n = 12$, 5.1%). A total of 146 studies (61.6%) did not report their validation level in sufficient detail for classification.

Only 16 studies (6.8%) reported prospective field validation—that is, testing of the AI-enhanced diagnostic tool in real-world conditions at the point of care in a resource-limited setting. This represents a pronounced 'valley of death' between proof-of-concept and deployment, consistent with broader observations about the translational gap in AI health technologies [17]. The prospective field studies span multiple diseases and modalities: Turbé et al. [10] validated an HIV RDT reader in the field with 11,374 tests; Weber et al. [33] conducted multi-modal TB/HIV screening on 541 patients; Lindblade et al. [32] evaluated malaria RDT reading in a field setting; Roche et al. [39] achieved 100% sensitivity and 99.1% specificity for HIV diagnosis on 190 patients; Guo et al. [36] validated a smartphone-based malaria microfluidic system; Prathap and Yadav [37] deployed YOLO-based malaria microscopy; Vu et al. [40] evaluated a logistic regression-based malaria and dengue screening tool on 485 patients; Kessler et al. [34] assessed pneumonia ultrasound on 604 patients; Brenes et al. [41] conducted prospective field validation of HPV cervical screening with CNN achieving 0.94 sensitivity and 0.94 specificity on 1600 images; Elbehiry et al. [42] validated an AMR detection system achieving 90.5% sensitivity and 99.6% specificity; and Afifi et al. [38] validated COVID-19/pneumonia chest x-ray interpretation on 1311 images. These 16 studies represent the evidence base most directly relevant to clinical implementation decisions.

The 12 studies using public datasets (such as the NIH malaria cell image dataset or Kaggle COVID-19 radiograph collections) raise additional concerns about generalizability. While benchmark datasets serve an important purpose in algorithm comparison and development, they cannot substitute for validation on

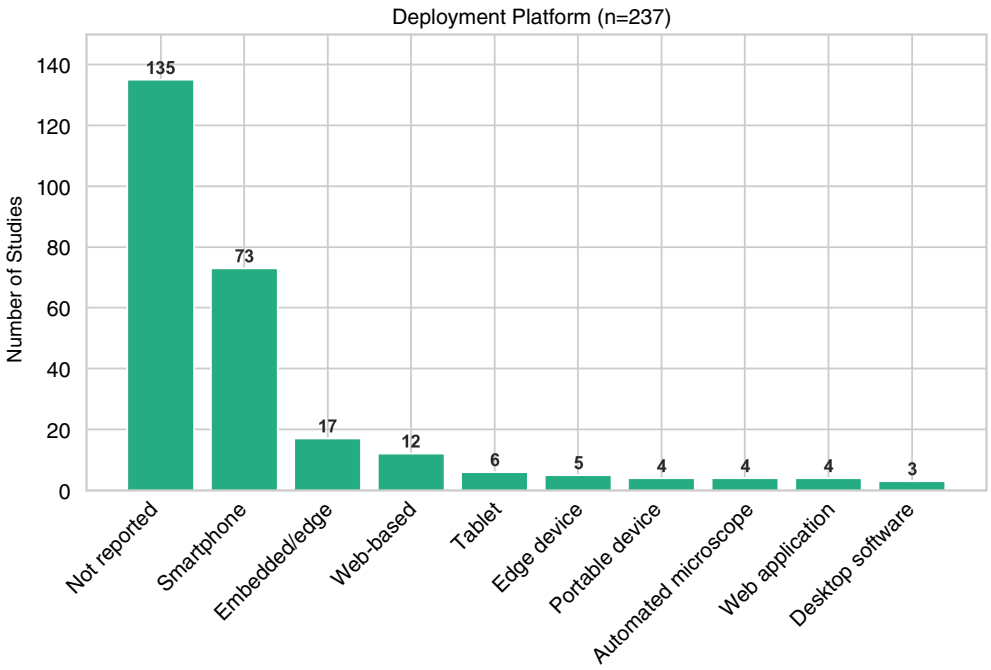


FIGURE 16 | Deployment platform distribution ($n = 237$). Smartphone-based applications are the most common reported platform; 135 studies (57%) did not report their deployment platform.

locally acquired data that reflect the specific conditions of the target deployment setting [43]. Public datasets may not represent the imaging conditions (lighting, camera quality, staining variability), disease prevalence, co-infection patterns, or population characteristics encountered in LMIC field settings. The study by Ho et al. [44], which used a public dataset of only 32 schistosomiasis images to evaluate a complex multi-architecture system (CNN, ResNet, Mask R-CNN, Transformer, Feature Pyramid Network), illustrates the risk of over-engineering relative to available evaluation data.

The validation gap was not uniform across diseases. A cross-tabulation of disease and validation level reveals that malaria studies had the highest proportion of prospective field validations, consistent with the longer history of malaria diagnostic research and the established pathways for field testing of malaria diagnostics in endemic settings. COVID-19 studies, despite their large numbers, were predominantly retrospective, reflecting the urgency-driven research context in which rapid publication was prioritised.

3.9 | Diagnostic Performance

Performance metrics were reported in 62 studies with varying completeness (Table 3). Sensitivity ranged from 83% to 100% across diseases and modalities (excluding biosensor studies reporting implausible values in non-standard units), while specificity ranged from 70% to 99.6%. Accuracy ranged from 81% to 100%, and AUC values ranged from 0.92 to 1.0 where reported. These figures should be interpreted with caution, as the majority were derived from retrospective or laboratory-based evaluations rather than prospective field studies. Publication bias toward high-performing models is likely, and direct comparison across studies is precluded by heterogeneity in reference standards, sample sizes and evaluation protocols.

Disease-specific performance patterns are informative when examined with appropriate caveats. For malaria microscopy, reported accuracies were consistently high, ranging from 94.81% [45] to 99.75% [37], with several studies reporting above 97% accuracy [46–50]. However, these studies varied markedly in sample size—from 282 specimens [51] to 27,558 [45] to 27,090 [47]—and validation level. Kone et al. [51], reporting a more modest 81.3% accuracy with 83.97% sensitivity and 80.92% specificity in a retrospective clinical validation of 282 specimens from Mali, may more accurately reflect real-world performance conditions.

For tuberculosis, chest x-ray-based AI tools showed sensitivity ranging from 84.1% [52] to 95.3% [Marquez, 2025], with specificity varying widely from 62.6% [53] to 97.7% [54]. This variability reflects the sensitivity–specificity trade-off inherent in screening applications and the influence of prevalence on predictive values. Weber et al. [33], in one of the few prospective field studies, reported balanced performance (95.0% sensitivity, 95.0% specificity) for TB/HIV screening using a multi-modal approach with random forest algorithms on 541 patients.

For COVID-19, performance reports were predominantly retrospective. Sagreiya et al. [55] reported 97.0% sensitivity and 93.0% accuracy for ultrasound-based COVID-19 detection using a multi-algorithm approach (CNN, U-Net, SVM, Decision Tree, LSTM). Rohaim et al. [56] reported 98.0% accuracy for molecular test-based COVID-19 detection using CNNs. Chandra et al. [57] achieved 87.0% sensitivity, 88.0% specificity and 89.5% accuracy for a biosensor-based approach.

For HIV, performance data were available from several studies spanning different modalities. Turbé et al. [10] reported the strongest field-validated evidence, with 95.9% sensitivity and 99.0% specificity for RDT interpretation in a prospective study of 11,374 tests. Roche et al. [39] achieved 100% sensitivity and 99.1% specificity in a smaller prospective field study ($n = 190$). Holmström et al. [58] reported 95.7% sensitivity and 98.5% specificity for HIV-related microscopy and imaging using CNNs on a public dataset. Bermejo-Peláez et al. [59] achieved an AUC of 0.997 for HIV and fungal infection RDT reading using MobileNet on 1272 samples in a laboratory setting.

Among the 19 studies reporting human-AI comparison, AI systems generally matched or exceeded expert performance and consistently outperformed non-expert health workers. This finding is encouraging for the use case of AI-enhanced diagnostics in settings with limited specialist personnel, though it requires confirmation through prospective field studies. The comparison is particularly relevant for malaria microscopy, where inter-reader variability among non-expert microscopists is a well-documented problem and for RDT interpretation, where subjective reading of faint test lines leads to both false-positive and false-negative results.

3.10 | Cost Data and Commercial Products

Cost data were sparsely reported across the included literature. The majority of studies did not provide any cost information, and those that did typically reported device costs or per-test costs without formal cost-effectiveness analysis. This represents a significant evidence gap, as cost is a primary determinant of adoption in resource-limited settings. The absence of cost data makes it difficult for policymakers and health system planners to compare AI-enhanced diagnostics against existing alternatives or to model the economic impact of deployment at scale.

Commercial product identification was similarly limited. The majority of studies described research prototypes rather than products with defined regulatory pathways or commercial availability. Among the exceptions, the Turbé et al. [10] HIV RDT reader and several TB CAD systems represent tools with more advanced commercial or deployment trajectories. The disconnect between the large number of published algorithms and the small number of commercially available or deployed products further illustrates the translational gap identified in the validation analysis.

3.11 | Implementation Barriers

Implementation barriers were reported in a minority of studies, typically in discussion sections rather than as primary

TABLE 3 | Diagnostic performance metrics of included studies reporting quantitative outcomes.

Study	Disease	Modality	Sens (%)	Spec (%)	Acc (%)	AUC	*N*	Validation
Elbehiry A 2025	AMR	—	90.5	99.6	—	0.96	—	Prospective field
Al-Zogbi L 2021	COVID-19	Ultrasound	—	—	20.6	—	19	Public dataset
Banga I 2024	COVID-19	Biosensor	—	91.1	89.1	—	46	Laboratory
Chandra M 2022	COVID-19	Biosensor	87	88	89.5	—	—	Retrospective clinical
Fakhry 2021	COVID-19	Audio/voice	—	—	—	0.99	—	—
Gomez-Gonzalez E 2022	COVID-19	Molecular/NAAT; Spectroscopy	100	87.5	87	0.95	72	Retrospective clinical
Kumar N 2022	COVID-19	Molecular/NAAT; Biosensor	11.1	—	100	—	—	Retrospective clinical
Kumar S 2023	COVID-19	RDT/LFA	95	—	—	—	—	Laboratory
Okila 2026	COVID-19	Ultrasound	—	—	95	—	—	—
Potter CJ 2022	COVID-19	Microscopy; RDT/LFA; Molecular/NAAT; Biosensor	—	—	81.3	—	128	Laboratory
Rohaim MA 2020	COVID-19	Molecular/NAAT	—	—	98	—	—	Retrospective clinical
Sagreiya H 2023	COVID-19	Ultrasound; Molecular/NAAT	97	—	93	0.97	—	Retrospective clinical
W. D. Puja 2024	COVID-19	Molecular/NAAT; Audio/voice	—	—	93	—	19	—
Afifi 2021	COVID-19; Pneumonia	Chest x-ray	—	—	91.2	—	1311	Prospective field
Murphy K 2023	COVID-19; Pneumonia	RDT/LFA; Chest x-ray; Molecular/NAAT	83.0	98.0	—	0.74	3231	Laboratory
Das S 2025	Dengue	Molecular/NAAT; Spectroscopy	—	—	—	1.00	302	Retrospective clinical
Mahmuda 2026	Dengue	—	—	—	88	—	170	—
Kolluru V 2025	Gonorrhoea; AMR	—	—	—	—	0.97	—	Public dataset
Holmström O 2021	HIV	Microscopy; Dermoscopy/ imaging	95.7	98.5	—	0.94	—	Public dataset
Jaiteh M 2025	HIV	—	97.8	100	—	—	—	Retrospective clinical
Roche SD 2024	HIV	—	100	99.1	—	—	190	Prospective field
S. Demir 2024	HIV	—	—	—	92	—	—	—
Turbé V 2021	HIV	RDT/LFA	95.9	99	—	—	11,374	Prospective field

(Continues)

TABLE 3 | (Continued)

Study	Disease	Modality	Sens (%)	Spec (%)	Acc (%)	AUC	*N*	Validation
Bermejo-Peláez D 2023	HIV; Fungal infections	RDT/LFA	—	—	—	1.00	1272	Laboratory
Brenes D 2022	HPV	Dermoscopy/imaging	94.0	94.0	—	—	1600	Prospective field
Xu T 2025	HPV	Molecular/NAAT; Microfluidic	10	—	94	—	10,976	Retrospective clinical
von Bahr J 2025	Helminth infections	Microscopy	61.1	98.7	—	—	777	Laboratory
Chen H 2025	Hepatitis B; Hepatitis C	Microfluidic	94.3	95.2	94.6	—	—	Retrospective clinical
Mahmud 2024	Hepatitis C	—	—	—	96.8	—	—	Public dataset
Choo H 2020	Influenza	—	84	80	82	—	—	Retrospective clinical
Appiah 2026	Malaria	Microscopy; RDT/LFA; Molecular/NAAT	38.9	60.6	—	—	—	—
Ashaolu JO 2025	Malaria	RDT/LFA	75.6	80.9	78.3	0.89	—	Retrospective clinical
Bahati 2025	Malaria	RDT/LFA	96.1	98	—	—	—	—
Frade S 2025	Malaria	RDT/LFA	92	97.3	—	—	—	Laboratory
Guo X 2021	Malaria	Microfluidic	—	—	97.8	—	—	Prospective field
Holmström O 2020	Malaria	Microscopy	89	—	—	—	—	Retrospective clinical
Hong SG 2025	Malaria	—	—	—	—	0.76	405	Retrospective clinical
Kone F 2025	Malaria	Microscopy	84.0	80.9	81.3	—	282	Retrospective clinical
Kumar 2022	Malaria	Microscopy	—	—	—	99.01	—	—
Laktabai J 2018	Malaria	RDT/LFA	—	93.9	—	—	—	Laboratory
Lebel P.M. 2025	Malaria	—	83	96	—	—	—	Laboratory
Lindblade KA 2025	Malaria	RDT/LFA	—	—	96.8	—	—	Prospective field
M. Mahajan 2024	Malaria	Microscopy	—	—	94.8	—	27,558	—
Morang'a CM 2020	Malaria	—	—	—	96.0	0.98	2207	Retrospective clinical
Ngufor C 2025	Malaria	RDT/LFA	—	—	96.8	—	—	Laboratory
Nithisha 2025	Malaria	Microscopy	—	—	99.7	—	—	—
Oktovianus 2024	Malaria	Microscopy	—	—	97.1	—	—	—
O'Connor 2025	Malaria	Microscopy	—	—	92	—	—	—

(Continues)

TABLE 3 | (Continued)

Study	Disease	Modality	Sens (%)	Spec (%)	Acc (%)	AUC	*N*	Validation
Parveen R 2025	Malaria	Microscopy; Molecular/NAAT	—	94.3	99.5	99.95	27,090	Public dataset
Penno E.C. 2015	Malaria	RDT/LFA	83.0	62.0	—	—	—	—
Polok 2025	Malaria	—	—	—	97	—	—	—
Prathap VM 2026	Malaria	Microscopy	—	—	99.8	—	—	Prospective field
Sharma 2025	Malaria	Microscopy	—	—	98.4	—	3925	—
Y. Zhong 2023	Malaria	Microscopy	—	—	97.7	—	—	—
Vu DM 2023	Malaria; Dengue	—	85.1	11	—	—	485	Prospective field
Kessler D. 2024	Pneumonia	Ultrasound	88	89	—	—	604	Prospective field; Retrospective clinical
Ho C 2025	Schistosomiasis	RDT/LFA	86.6	98.5	—	—	32	Public dataset
Jøker KE 2026	Schistosomiasis	—	84.0	81.0	81.0	0.90	583	Laboratory
Meulah B 2024	Schistosomiasis	Microscopy; RDT/LFA; Molecular/NAAT	83.1	—	—	—	798	Retrospective clinical
Chattoraj S. 2025	Tuberculosis	Chest x-ray	84.1	80.6	—	—	10,000	Retrospective clinical
Karlita 2024	Tuberculosis	Microscopy	—	—	97.6	—	—	—
Khan 2025	Tuberculosis	Chest x-ray	—	97.7	95.5	0.98	—	—
Koeppel L 2023	Tuberculosis	—	90	70	—	0.92	—	Laboratory
Marquez N 2025	Tuberculosis	Chest x-ray	95.3	62.6	—	1.00	—	Retrospective clinical
Wei X 2025	Tuberculosis	Chest x-ray; Molecular/NAAT	86	—	—	—	—	Laboratory
Yang Z 2025	Tuberculosis	Molecular/NAAT	97	98.1	95.2	1.00	—	Laboratory
Shankar 2025	Tuberculosis; COVID-19; Pneumonia; Respiratory infections	Ultrasound	—	—	98.1	—	—	Public dataset
Majam M 2023	Tuberculosis; HIV	—	84	71	—	82.84	593	Retrospective clinical
Weber SF 2025	Tuberculosis; HIV	Chest x-ray; Ultrasound; Molecular/NAAT	95	95	90	0.82	541	Prospective field
Wu C 2026	Tuberculosis; HIV	Chest x-ray	74.1	84	—	0.95	130	Laboratory
Berg B 2015	—	—	—	—	99.6	—	—	—

(Continues)

TABLE 3 | (Continued)

Study	Disease	Modality	Sens (%)	Spec (%)	Acc (%)	AUC	*N*	Validation
Manyazewal T 2024	—	—	—	100	87	—	—	Laboratory
V. Swetha 2025	—	—	—	—	93.5	—	—	—

Note: Values reported as proportions (0–1) in original studies have been converted to percentages. Studies reporting implausible values in non-standard units (primarily biosensor studies reporting in analyte concentration rather than diagnostic accuracy) have been excluded. Em-dash (—) indicates not reported.
Abbreviations: Acc, accuracy; AUC, area under the receiver operating characteristic curve; *N*, sample size; Sens, sensitivity; Spec, specificity.

research findings. Where reported, barriers fell into several thematic categories. Infrastructure barriers included unreliable electricity supply, limited or absent internet connectivity and inadequate cold chain or storage facilities. These are particularly relevant for cloud-based AI systems that require consistent connectivity. Human resource barriers included the need for training health workers to use AI-enhanced tools, resistance to technology adoption and concerns about trust in AI-generated results. Regulatory barriers included the absence of clear regulatory pathways for AI-enhanced diagnostics in many LMICs, the complexity of navigating multiple national regulatory frameworks for multi-country deployment and the lack of WHO prequalification pathways specifically designed for AI-enhanced diagnostic tools. Data-related barriers included the scarcity of labelled training data from LMIC settings, concerns about data privacy and sovereignty and the challenge of maintaining model performance across diverse populations and settings. Economic barriers included device costs, recurring costs for cloud processing and the absence of reimbursement mechanisms for AI-enhanced diagnostics in LMIC health financing systems.

The limited and unsystematic reporting of implementation barriers across the literature reflects the predominant focus of included studies on algorithm development and technical performance rather than on the practical challenges of deployment. This is consistent with the overall finding that the field remains concentrated at the pre-deployment stage.

3.12 | Post Hoc Explanatory Cross-Tabulations

3.12.1 | Validation Maturity by Disease and Income Class

The 91 studies with extractable validation level were cross-tabulated against disease and against World Bank income class (Figures 7 and 8). The disease cross-tabulation (Figure 7) shows that the better-represented diseases concentrate at different points of the validation pipeline: COVID-19 studies cluster at the laboratory and retrospective stages (10 of 24 with validation data at laboratory; 9 at retrospective; only 1 prospective field), malaria distributes more evenly (8 laboratory, 7 retrospective, 6 prospective field of 24), HIV achieves the highest prospective-field share of the larger disease groups (4 of 12 with validation data) and tuberculosis remains predominantly at laboratory and retrospective stages (5 and 3 of 11).

The income-class cross-tabulation (Figure 8) does *not* support a simple 'HIC develops, LMIC tests' pattern within the included literature. Of studies reporting both income class and validation level, HIC-affiliated studies ($n = 8$) are concentrated at later validation stages (75% retrospective or prospective; 25% prospective field), UMIC-affiliated studies ($n = 7$) report 14% prospective field, LMIC-affiliated studies ($n = 24$) report 21% prospective field and LIC-affiliated studies ($n = 9$) report the highest prospective-field share at 44%. Sample sizes per income class are small and the corpus has 71% missingness on income class, so this pattern is reported as a structural observation rather than a statistical claim. The data are nevertheless inconsistent with the assumption that HIC-developed tools dominate

the early-stage literature while LMIC partners conduct field validation; the relationship is closer to 'wherever a field-deployable research network already exists, field validation follows' (cf. the Mali malaria research partnership and the East African HIV diagnostic networks).

3.12.2 | Reporting-Completeness Sensitivity Analysis

Across the 26 charting variables, the three core operational variables flagged by peer review—country (39% reported), deployment platform (43% reported) and validation level (38% reported)—combine to yield a complete-case subset of only 40 of 237 studies (17%). The remaining 197 studies (83%) are missing at least one of the three (Figure 9). Disease distribution differs systematically between the two subsets (Figure 10): in the complete-case subset, HIV is over-represented (17% vs. 5% in the missing-case subset; +12 percentage points), Tuberculosis is under-represented (2% vs. 14%; −12 percentage points) and COVID-19 is under-represented (21% vs. 30%; −8 percentage points). AI-technique distribution differs less (maximum absolute difference 6 percentage points; Figure 11), suggesting that the field's overall technique-dominance picture is robust to reporting selection while the disease-coverage picture is partly an artefact of which studies report their data well. We treat this as a substantive caveat on Section 3.3 disease counts rather than as evidence that the disease counts are misleading; the direction of the bias is toward over-reporting in HIV and under-reporting in TB.

3.12.3 | Architecture × Validation: Top 50 by Citation

OpenAlex citation counts were retrieved for 235 of 237 studies (99%). Top 50 by citation span 28–1073 citations (median 48). The top 50 contained no studies using explicitly lightweight architectures (MobileNet, YOLO-tiny, SqueezeNet, EfficientNet-Lite, MobileViT, ShuffleNet); the top 50 is composed of 17 generic-CNN/deep-learning studies (untyped), 15 architecture-unreported studies, 12 non-deep-learning/classical-ML studies, 4 heavy-architecture studies (ResNet, EfficientNet, Vision Transformer) and 2 medium-architecture studies (Figure 12). Among the top 50, prospective field validation is reached only by generic-CNN studies (5 of 17, 29%); the 4 heavy-architecture and 2 medium-architecture top-50 papers all remain at laboratory, retrospective clinical, or unreported validation stages.

In the full-corpus robustness check (235 studies with citation data), lightweight architectures appear in only 3 studies but reach prospective field validation in 33% of them (1 of 3); heavy-architecture studies appear in 21 cases and reach prospective field validation in 0% of them. The literature rewards architectural complexity with citations and penalises it on deployment evidence. This is the structural finding the Section 4.1 framing builds on.

3.12.4 | Cost-Data Reporting × Validation Level

Cost data of any kind (per-test cost, hardware cost, comparator cost) were reported in 61 of 237 studies (26%). The reporting rate is essentially flat across validation maturity (Figure 13): laboratory studies report cost in 37% of cases (13 of 35); prospective field studies report cost in 38% of cases (6 of 16). Cost reporting does not mature with the validation pipeline. The 10 of 16 prospective-field studies that do not report cost include several with otherwise complete charting profiles, indicating that the omission is not a function of insufficient publication length but of an unreported convention.

4 | Discussion

4.1 | Principal Findings

This scoping review maps 237 studies of AI-enhanced point-of-care diagnostics for infectious diseases in resource-limited settings, published between 2015 and early 2026, and asks why a field producing studies at an exponential pace converts so few of them into prospective field-validated tools. The headline numbers describe the magnitude of the problem: only 16 of 237 studies (6.8%) report prospective field validation, while 62% do not report a validation level at all, and only 26% report any cost information. The more useful contribution of this review is the *structural pattern* that explains where the conversion fails. Four findings (Section 3.12), taken together, suggest the field is locked in a high-output/low-conversion mode rather than a high-output/high-conversion one.

First, validation maturity is unevenly distributed across diseases, and the pattern is not the one the literature has tended to assume. Of the 91 studies with extractable validation level, HIC-affiliated studies ($n = 8$) are more concentrated at late-stage validation (75% retrospective or prospective) than LMIC-affiliated studies ($n = 24$, 50% late-stage). Low-income-country-affiliated studies ($n = 9$) report the highest prospective-field share (44%). The simple 'HIC develops, LMIC tests' framing does not hold in this corpus; the relationship is closer to 'wherever a field-deployable research network already exists, field validation follows'.

Second, architectural choice and citation success move in opposite directions to deployment success (Section 3.12.3). The top 50 studies by OpenAlex citation count contain no lightweight architectures (MobileNet, YOLO-tiny, SqueezeNet, EfficientNet-Lite) at all, and the 4 top-50 studies using heavy architectures (ResNet, EfficientNet, Vision Transformer family) achieve 0% prospective field validation. Across the full corpus, lightweight architectures appear in only 3 studies—yet one in three of those (33%) reach prospective field validation, compared with 0 of 21 heavy-architecture studies. The literature *rewards* architectural complexity with citations and *penalises* it on deployment.

Third, cost reporting does not mature with the validation pipeline (Section 3.12.4). Studies at the laboratory stage report cost in 37% of cases; studies that progress to prospective field validation report cost in 38% of cases. The transition from bench to field is not accompanied by an increase in the policy-relevant information needed to evaluate adoption.

Fourth, the descriptive summaries are themselves partly an artefact of who reports their data well (Section 3.12.2). Only 40 of 237 studies (17%) are complete on the three core charting variables

(country, deployment platform, validation level). Among the 40 complete-case studies, HIV is over-represented (17% vs. 5% in the missing-case subset) and tuberculosis under-represented (2% vs. 14%), a 12-percentage-point gap in each direction. Disease-coverage figures throughout the literature are therefore visible only through the lens of reporting completeness.

Six principal findings emerged from the descriptive synthesis, summarised below; the structural pattern just described connects them. The field is growing exponentially: 44% of studies were published in 2025–2026 alone. Disease coverage is concentrated on three conditions—COVID-19, malaria and tuberculosis—which together account for 73% of the corpus. Only 6.8% of studies report prospective field validation, revealing a pronounced translational gap. Geographic representation clusters in East Africa and South Asia, with West Africa, Central Africa, Latin America and the Pacific significantly under-represented. CNNs dominate the technique mix, with limited adoption of newer architectures. Implementation barriers, cost data and deployment platform information are systematically under-reported across the corpus.

4.2 | Exponential Growth and the COVID-19 Effect

The exponential growth trajectory, with 44% of included studies published in 2025–2026, reflects both the maturation of deep learning frameworks and the catalytic effect of the COVID-19 pandemic. The pandemic drove urgent investment in decentralised diagnostic solutions and created a natural laboratory for testing AI-enhanced POC tools at scale [60]. The surge in COVID-19 diagnostic research was not limited to molecular testing: it stimulated innovation across multiple modalities including chest imaging, cough-based audio screening, biosensors and antigen test interpretation, several of which have applicability beyond COVID-19. However, this growth has been unevenly distributed: COVID-19 alone accounts for 32% of all included studies, potentially distorting the field's development away from diseases with the greatest burden in resource-limited settings, such as malaria, TB and neglected tropical diseases. Whether the infrastructure and funding momentum generated by COVID-19 diagnostic research will be redirected toward high-burden endemic diseases remains to be seen.

The publication trajectory also raises questions about the maturity and quality of the evidence base. The concentration of nearly half of all studies within the most recent 18 months means that much of the literature has not yet been subjected to independent replication, external validation, or the scrutiny that comes with extended use in the field. Rapid growth in publications does not necessarily equate to rapid progress in clinical implementation.

4.3 | Disease Coverage Inequity

The dominance of COVID-19, malaria and TB—collectively representing 73% of disease-specific studies—leaves significant gaps for other infectious diseases of major public health importance in LMICs. Neglected tropical diseases (NTDs), which affect over 1.7 billion people globally [61], account for fewer than 8% of included studies. The NTDs that are represented—schistosomiasis ($n = 7$), dengue ($n = 7$), leishmaniasis ($n = 3$) and helminth infections ($n = 4$)—are themselves skewed toward diseases where image-based diagnosis (microscopy) is an established pathway. Diseases diagnosed primarily through clinical algorithms, serological testing, or molecular methods have attracted less AI innovation.

Sexually transmitted infections other than HIV are nearly absent from the literature. Gonorrhoea ($n = 2$), HPV ($n = 3$) and syphilis received minimal attention despite their substantial burden in LMICs and the clear potential for AI-enhanced POC testing—for example, automated interpretation of syphilis rapid tests at antenatal care visits. Antimicrobial resistance, a cross-cutting public health emergency, appeared in only 8 studies despite the clear need for rapid, POC-level AMR detection to guide empiric therapy in settings without access to culture-based susceptibility testing.

This pattern mirrors broader inequities in global health research funding and attention [62], suggesting that AI diagnostic development is reproducing rather than disrupting existing biases in the research landscape. Diseases with established research networks, large public datasets and well-defined imaging workflows (malaria blood smears, TB chest x-rays) attract disproportionate AI research attention, while diseases lacking these preconditions remain underserved regardless of their public health burden.

4.4 | The Validation Valley of Death

Prospective field validation is the exception rather than the rule in this corpus: just over one in 15 studies (16 of 237) reaches it—a finding with direct implications for clinical decision-making and policy. The overwhelming majority of AI-enhanced POC diagnostic tools remain at the laboratory proof-of-concept or retrospective evaluation stage. This 'valley of death' is well-documented in digital health more broadly [17, 43], but its magnitude in this specific domain has not been previously quantified. The validation pipeline (Figure 5) illustrates the steep attrition: of the 91 studies with extractable validation data, only 16 progressed to prospective field testing, and none reported sustained real-world deployment or integration into routine health systems.

Several factors contribute to this translational gap. Prospective field studies require regulatory approvals, ethical review in multiple jurisdictions, partnerships with local health systems and sustained funding beyond the typical grant cycle for algorithm development. The technical challenges of deploying AI models in real-world conditions—variable imaging quality, intermittent connectivity, device heterogeneity and different disease prevalence from training populations—further complicate the transition from laboratory to field [63]. Additionally, many AI researchers lack the implementation science expertise, local partnerships and health systems knowledge needed to design and execute field validation studies, while LMIC-based researchers who possess this knowledge may lack the computational resources and AI expertise to develop the tools.

The high proportion of studies with unreported validation levels (62%) is itself a concern, suggesting that many studies either did

not clearly define their validation approach or conducted evaluations that resist straightforward classification. This finding strengthens the case for domain-specific reporting guidelines (discussed in Section 4.7).

The high proportion of studies using public datasets (5.1%) is a related concern. While benchmark datasets serve an important purpose in algorithm comparison and development, they cannot substitute for validation on locally acquired data that reflect the specific conditions of the target deployment setting [43]. The reliance on curated datasets from the NIH, Kaggle and similar repositories introduces a particular risk: models that perform well on these datasets may fail when confronted with the variability of real-world imaging conditions in LMIC settings, including different camera models, lighting conditions, staining protocols and patient populations.

The validation discussion above presumes that disagreement between an AI tool and a human reference reader is *evidence against the tool*. Nineteen studies in the corpus (8%) report a direct comparison with human-expert interpretation as the reference standard; these include Holmström et al. [64] for HIV/microscopy, von Bahr et al. [20] for helminth detection by digital microscopy, Meulah et al. [15] for schistosomiasis and Brenes et al. [26] for HPV/cervical cytology. The implicit framing is that an AI-vs-expert delta is a tool failure. This framing is incomplete in two ways. First, human expert agreement in the relevant tasks—parasite egg identification on Kato-Katz smears, weak-positive line reading on lateral-flow tests, smear microscopy for tuberculosis, dermatological surface inspection—is itself variable and well below 100%; reader variability is the established reason POC tools are sought in the first place [19, 65]. Treating expert reads as ground truth in such tasks lowers the ceiling on apparent AI accuracy and obscures both directions of disagreement. Second, where AI disagrees with experts, the disagreement may itself be informative: detection of low-density malaria parasites missed by human microscopists; identification of faint RDT bands at the limit of visual interpretation; or pattern recognition in chest-radiograph subtleties that experts inconsistently see. We recommend that future field-validation studies in this space report AI-vs-expert agreement against a higher-tier reference standard where one exists (PCR or culture for infectious disease diagnosis; histopathology for malignancy) and use AI–expert disagreement *both* as a source of error analysis and as a candidate signal for pattern discovery.

A related under-engaged dimension is failure-mode analysis: which samples does the model get wrong, and why? Of the 237 included studies, only 43 (18%) report any qualitative analysis of error modes in our extraction, and these cluster into a small number of categories: connectivity disruption (15 studies report explicit failure or degraded performance under offline or intermittent-network conditions—e.g., Turbé et al. [10], Allan-Blitz et al. [66], Hou et al. [67]); generalisability shifts under prevalence or population change (5 studies—e.g., Parveen et al. on malaria smartphone microscopy, Bao et al. on COVID-19, Colubri et al. [68] on Ebola, Weber et al. on TB/HIV co-infection); power and infrastructure failure (5 studies—Choudhary et al., Hardie et al., Madukwe et al.); data-quality failures including motion blur, low resolution, debris on slides and operator variability in sample preparation (Ngufor et al. for malaria; Marquez et al. for TB); and regulatory/governance failures that prevented deployment despite acceptable accuracy (Chen et al., Li et al., Yadav et al., Berg et al.). The remaining 82% of studies report aggregate accuracy figures with no decomposition by error mode. This is a substantive reporting gap: a tool that achieves 95% sensitivity on average but loses 10 percentage points on the lowest-quality 10% of images is not deployable in a setting where the lowest-quality images are precisely the ones that will arrive in routine practice. The POC-AI-LMIC reporting extension proposed in Section 4.7 (M2) accordingly mandates failure-mode categorisation across five axes: image quality, sample-preparation variability, prevalence shift, infrastructure dependency and operator inter-rater agreement.

4.4.1 | Two Studies in Detail

The Turbé et al. [10] HIV rapid-diagnostic-test study exemplifies the design pattern this corpus generally lacks. The team trained a deep-learning classifier on 11,374 lateral-flow assay images (CNN + SVM ensemble) collected across multiple South African community-testing sites, deliberately spanning the visual variability that distinguishes routine deployment from a laboratory benchmark: faint and bright reaction lines, multiple HIV antibody RDT brands, variable lighting, smartphone camera classes and operator-controlled image framing. The reference standard combined the manufacturer's RDT reading, ELISA confirmation and expert clinician interpretation—a higher-tier reference than the per-test visual read alone. The deployment context was a Cape Town community-testing site with smartphone and tablet hardware available at the point of care; prospective field validation reported a sensitivity of 95.9% and specificity of 99.0%, with explicit reporting of cost, connectivity barriers (which the authors flagged as a limitation rather than concealing) and operator-acceptability data. The study reports a complete charting profile across the dimensions this review extracted—country, income class, setting type, training data size, reference standard, performance metrics, validation level, cost data and implementation barriers—making it one of the few in the corpus that supports a downstream policy decision. The specific design features worth replicating: training-set size large enough to span deployment variability; multiple-tier reference standard with explicit clinical concordance; community-setting prospective field validation rather than laboratory-only evaluation; transparent reporting of barriers (including the connectivity dependency) rather than a clean-data success narrative; and disclosure of cost data sufficient for a cost-effectiveness comparator. Turbé et al. did not produce a perfect tool, but they produced one that a regulator or a programme manager can evaluate.

The Ho et al. [44] preprint illustrates a recurring failure mode at the opposite end. The study evaluates an ensemble of CNN, ResNet, Mask R-CNN, Transformer and Feature Pyramid Network components for AI-assisted interpretation of schistosomiasis lateral-flow rapid diagnostic tests on tablet hardware in a field setting in Uganda, and reports 86.6% sensitivity and 98.5% specificity. Headline performance is in the same range as Turbé et al. and the deployment-context narrative is consistent with the kind of LIC field-relevant study the literature is

short on. The substantive concern is the training data: only 32 images were used to train the multi-architecture ensemble, and the reported validation was on a public dataset rather than on a held-out subset of the field data the team collected. The five architectures span heavy and computationally expensive families; the combination of a 32-image training set with a multi-component heavy ensemble is statistically incompatible with the generalisation claims supported by the reported performance. The reported sensitivity and specificity are most likely sample-specific artefacts of an under-powered training run validated on a benchmark that does not represent the deployment population, rather than evidence of a deployable schistosomiasis screening tool. The study illustrates three patterns this review identifies as endemic: (i) high-performance numbers reported in a venue (preprint) without complete pre-registration of a validation plan; (ii) architecture choices over-fit to a small training corpus, with a public-dataset validation step that does not substitute for held-out field data and (iii) absence of reference-standard tiering and failure-mode reporting that would let a reviewer evaluate the generalisation claim. The contrast with Turbé et al. is not in the disease, the setting, or the deployment hardware—it is in the alignment between the training-data scale, the architectural complexity and the validation evidence presented.

4.5 | Geographic Gaps and the HIC Development Paradox

The geographic distribution of studies reveals a paradox: while the tools under development are intended for resource-limited settings, a substantial proportion of research originates from high-income country laboratories. Only 39% of studies had extractable country-level data, and among these, HIC-based institutions feature prominently (USA $n = 10$, Switzerland $n = 8$, UK $n = 6$). This pattern is not inherently problematic—HIC institutions often have the computational resources and technical expertise for algorithm development—but it raises questions about the involvement of end-user communities in tool design and validation. Tools developed without input from the health workers, patients and health system administrators who will use them in LMIC settings are at risk of addressing the wrong problems, making incorrect assumptions about infrastructure and workflow and failing to achieve adoption [63]. The Section 3.12.1 cross-tabulation also indicates that the *direction* of this paradox is not the conventional one within the included literature: HIC-affiliated studies in the corpus skew toward later-stage (retrospective or prospective) validation rather than earlier-stage development and LIC-affiliated studies ($n = 9$ with extractable income data) reach prospective field validation more often (44%) than HIC- or LMIC-affiliated studies. The empirically supported framing is therefore not 'HIC develops, LMIC tests' but 'wherever a field-deployable research network already exists, field validation follows'—with the Mali malaria network, the East African HIV testing programmes and the Cape Town RDT-reader collaboration as the strongest exemplars. Geographic equity in this field requires investment in *network capacity*, not only in algorithm development.

The concentration of LMIC-based studies in East Africa (Uganda, Kenya, Tanzania) and South Asia (India, Bangladesh) creates its own equity concern: West African countries, Central Africa, Central Asia and the Pacific Islands are significantly underrepresented despite bearing substantial infectious disease burdens. This gap may reflect research infrastructure and funding networks rather than need [69]. The presence of Mali ($n = 8$) as a well-represented West African country illustrates the role of established research partnerships—in this case, long-standing malaria research collaborations—in driving AI diagnostic research in specific locations. Countries without such pre-existing partnerships may be systematically excluded from the AI diagnostic research agenda regardless of disease burden.

The absence of Latin American representation beyond Brazil and Peru is particularly notable given the region's burden of dengue, Chagas disease, leishmaniasis and Zika virus, all of which are amenable to POC diagnostic approaches. The inaccessibility of the Global Index Medicus, which aggregates LILACS (the primary index for Latin American health literature), may have contributed to the underrepresentation of Latin American studies in this review, as discussed in the limitations section.

4.6 | Technical Landscape: CNN Dominance and Emerging Approaches

The dominance of CNNs (26% of studies) is expected given the predominance of image-based diagnostic tasks. However, the reliance on established architectures (ResNet, VGG, Inception) and the limited adoption of newer approaches (vision transformers $n = 3$, federated learning $n = 1$, self-supervised learning absent) suggest that the field may benefit from greater engagement with state-of-the-art machine learning research. The mean lag between the publication of a novel architecture in the machine learning literature and its application in AI-enhanced POC diagnostics appears to be several years, potentially limiting the field's ability to leverage the most efficient and capable models.

Federated learning, in particular, offers a promising solution for training models across distributed LMIC sites without centralising sensitive patient data [70]. Given the well-documented challenges of data scarcity and data sovereignty in LMIC health research, the near-absence of federated learning from this literature is a missed opportunity. Similarly, self-supervised and semi-supervised learning approaches, which can leverage large quantities of unlabelled data to improve model performance, are directly relevant to the LMIC context where labelled medical data are scarce but unlabelled data from routine diagnostic workflows may be abundant.

The trade-off between model complexity and field robustness is also under-engaged with in the literature. Heavy architectures benchmark well on curated datasets and accumulate citations (Section 3.12.3), but the corpus contains no instance of a heavy architecture (ResNet, EfficientNet B-series, Vision Transformer family) reaching prospective field validation; the three studies in the corpus using explicitly lightweight architectures—Allan-Blitz et al. [66] for gonorrhoea RDT interpretation on smartphone using a CNN/MobileNet/YOLO stack in a prospective field deployment; Bermejo-Peláez et al. for HIV/fungal-infection multiplex assay with MobileNet on smartphone

and web-based deployment at laboratory stage; and Guel et al. for malaria microscopy with a ResNet/VGG/MobileNet/YOLO ensemble on smartphone in early-stage development—all chose lightweight or hybrid architectures for the same operational reason: smartphone-class hardware, intermittent connectivity and unpredictable image-acquisition conditions impose constraints that a heavy architecture cannot meet without an external compute path. The implication is that complexity gains visible in benchmark accuracy can mask robustness losses that only emerge under the deployment conditions the tool was designed for—variable lighting, sample-preparation noise, low-end smartphone cameras, intermittent or absent connectivity and patient populations that may differ in disease prevalence and presentation from the training population. Field-relevant robustness—performance under domain shift, on low-quality images and on devices outside the development laboratory's reference set—is rarely reported in the corpus, and where it is reported the most-cited papers do not appear. A study's headline accuracy should be read alongside the *operational envelope* it was evaluated within: dataset provenance, hardware class, connectivity assumption and image-quality range. Without those, accuracy is uncomparable across studies and unable to inform deployment decisions in the settings for which the tool is built.

Transfer learning from ImageNet-pretrained models, reported in 10 studies, addresses the common challenge of limited labelled medical imaging data in LMIC settings but introduces domain gap concerns that require careful evaluation [71]. The substantial difference between natural images (on which ImageNet models are trained) and medical images (microscopy, radiographs, RDT photographs) means that transfer learning benefits are not guaranteed and may vary by modality and disease. The relatively low sensitivity reported by von Bahr et al. [20] for helminth detection using transfer learning (61.1%) exemplifies this concern.

The continued use of classical machine learning algorithms (random forests, SVMs, logistic regression, decision trees) alongside deep learning is a notable feature of this field. In resource-constrained settings, these algorithms offer practical advantages: they are computationally lightweight, can run on basic hardware, are more interpretable (an important consideration for clinical trust and regulatory approval), and require smaller training datasets. The diversity of AI techniques is therefore not a limitation but rather reflects appropriate adaptation to the constraints of the deployment context.

4.7 | Implications for Policy and Practice

The findings of this review have implications for three distinct layers of the research and deployment ecosystem. Each layer requires different actors and different time-scales. We frame six actionable recommendations stratified across *technical*, *methodological* and *systemic* tiers; the empirical basis for each is given in parentheses with the relevant Section 3 result reference.

Technical recommendations. T1—Match architectural complexity to deployment context, and report this choice as a design decision. Heavy architectures (ResNet, EfficientNet, Vision Transformer family) accumulate disproportionate citations in this corpus but are absent from the prospective field-validated subset (0 of 21; Section 3.12.3). Lightweight architectures (MobileNet, YOLO-tiny, SqueezeNet, EfficientNet-Lite) appear in only three studies but reach prospective field validation in 33% of cases. Algorithm developers and reviewers should treat the choice of a heavy architecture for a smartphone-deployed POC application as a justifiable but not default decision, and journals should require explicit reporting of the trade-off between accuracy on benchmark data and inference cost on the target hardware. *T2—Establish federated and privacy-preserving training infrastructure for LMIC POC contexts*. Federated learning was present in only one included study and self-supervised learning was absent (Section 3.5), despite both being directly relevant to settings characterised by abundant unlabelled diagnostic data and constrained data-sharing capacity. Multi-country consortia (building on existing networks such as the African Society for Laboratory Medicine (ASLM) and the Asia-Pacific Society for Infection Control) should host shared model registries and federated training nodes so that diagnostic AI can be improved without exporting patient data. The FIND Data Standard Framework for AI-based Diagnostics [72] provides an operational starting point.

Methodological recommendations. M1—Tie funding to a pre-registered, multi-site prospective field validation plan with explicit cost reporting. Of 237 included studies, only 16 (6.8%) report prospective field validation (Section 3.8) and only 38% of those report any cost data (Section 3.12.4), a rate indistinguishable from the 37% reported by laboratory-stage studies. Current funding mechanisms reward algorithm novelty rather than deployment readiness. Funders such as the Global Fund, FIND, WHO/TDR, NIH Fogarty, the Wellcome Trust and bilateral development agencies should require a pre-registered prospective field validation plan with explicit cost-per-test and infrastructure-cost reporting as a condition of POC AI diagnostic development grants beyond the proof-of-concept stage, following the precedent of the Global Fund × Siemens Healthineers AI-CAD scale-up partnership [73]. *M2—Adopt a domain-specific reporting extension for AI-enhanced POC diagnostics, modelled on STARD-AI and TRIPOD + AI*. Validation level is unreported in 62% of studies, deployment platform in 57%, and country of conduct in 61% (Section 3 throughout). STARD-AI [74] and TRIPOD+AI [75] address diagnostic-accuracy and prediction-model reporting respectively, but neither captures the LMIC-specific dimensions (target user, hardware/connectivity envelope, training-data provenance, cost-per-test, deployment partner). A POC-AI-LMIC reporting checklist should be developed under the EQUATOR Network [76] and mandated by journals publishing in this space, with required fields covering: target user profile (community health worker, laboratory technician, clinician), connectivity requirement (offline-capable, intermittent, always-online), hardware envelope (smartphone class, edge device, workstation), training-data origin (HIC, LMIC, mixed) and prevalence shift assessment, cost-per-test and cost of infrastructure and end-user-involvement statement.

Systemic recommendations. S1—Establish a WHO prequalification/regulatory pathway for AI-enhanced diagnostics and invest in LMIC computational capacity in parallel. AI POC diagnostics currently occupy a regulatory grey zone between medical-device

and software regulation, contributing to deployment uncertainty for funders, regulators and health system administrators. WHO and partner regulators should formalise an AI-enhanced diagnostic prequalification track, extending the existing WHO prequalification programme for diagnostics [77] and drawing on the WHO 2024 LMM ethics and governance guidance [78] and the WHO 2021 AI for health guidance [79]. In parallel, programmes such as the Lacuna Fund and the Africa Centres for Disease Control's pathogen genomics initiative should be extended to fund LMIC-resident computational capacity, datasets and research leadership, addressing the technical bottleneck that currently constrains LMIC-led AI diagnostic development. *S2—Issue targeted equity-driven funding calls covering both disease coverage and end-user co-design*. COVID-19, malaria and tuberculosis account for 73% of the corpus (Section 3.3); neglected tropical diseases, sexually transmitted infections (other than HIV) and antimicrobial resistance are systematically under-represented despite substantial LMIC disease burden. Funding calls should ring-fence allocations for under-represented disease–region pairs (e.g., dengue and Chagas in Latin America; soil-transmitted helminths and onchocerciasis in West Africa; gonorrhoea, syphilis and HPV across LMIC primary care) and require human-centred co-design with community health workers, laboratory technicians and end-users from the design stage onward, rather than at the field-validation stage [63].

4.7.1 | Equity, Governance and the Relevance of Non-AI Alternatives

The equity case for AI-enhanced POC diagnostics in LMICs depends on three assumptions that are not automatically satisfied: that AI tools will be available where they are needed, that they will perform fairly across LMIC patient populations, and that they will not displace simpler alternatives that are already deployable. None of these can be taken for granted from the current evidence base. Training data in the corpus we map are predominantly HIC-derived or curated from public benchmark sets [64, 80]; the under-representation of LMIC patient demographics, disease presentations, sample-preparation conditions and device variability introduces a documented risk of unequal performance at deployment [78, 79]. The data-governance environment is also asymmetric: many LMICs lack the consent infrastructure, data-protection regulation and sovereign data-stewardship capacity that LMM-era guidance assumes [78] and the dominant federated and privacy-preserving training architectures that would mitigate this risk (Section 4.6/T2) are virtually absent from the corpus. Compounding these technical and governance risks, the infrastructure dependencies of AI POC diagnostics—smartphone class, charging capacity, intermittent connectivity, app-store mediation, model-update cadence—can themselves widen rather than narrow access gaps if deployment is left to market mechanisms rather than coordinated by public health systems [19, 73]. The equity case for AI-enhanced POC therefore depends on regulatory action (S1), reporting transparency (M2) and architectural restraint (T1), not on diagnostic-accuracy improvements alone.

The case is further conditioned by the existence of non-AI rapid-diagnostic technologies that are already deployable. CRISPR-based visual-readout platforms (SHERLOCK [81], DETECTR [82]) provide nucleic-acid-level specificity in a lateral-flow format that can be read with the naked eye at room temperature, with field implementations reported for Zika and dengue [83]. Isothermal nucleic-acid amplification—loop-mediated isothermal amplification (LAMP) and recombinase polymerase amplification (RPA)—eliminates the thermal-cycler hardware burden of PCR and has been deployed for malaria, HIV and febrile-illness panels in community-health-worker contexts [84, 85]. Microfluidic and paper-based colourimetric assays, built on the Whitesides paradigm [86], allow multiplex visual readouts on sub-dollar substrates with no on-board electronics. These technologies do not require image-classifier interpretation, do not generate training-data demands, do not depend on smartphone hardware turnover and have established regulatory and quality-assurance pathways at FIND, WHO and the Global Fund. They have weaknesses—limited multiplexing relative to image-based AI; technician-variable interpretation at faint endpoints; quality-control challenges at scale—but those weaknesses are well characterised and addressable. Any responsible reading of 'more field validation is needed' for AI-enhanced POC diagnostics must include comparison with the relevant deployable non-AI alternative, not only against centralised laboratory reference standards. We recommend that future field-validation studies in this space include a non-AI rapid-diagnostic comparator and report cost-per-test, time-to-result and operator-training requirement on the same axes for both arms.

4.8 | Comparison With Adjacent Reviews

This review's findings align with and extend those of adjacent reviews. Schwalbe and Wahl [18] identified a broad gap between AI's potential and its realisation in global health, without quantifying the specific translational gap for POC diagnostics. Wahl et al. [19] discussed AI's potential in resource-poor settings, a theme this review extends with quantitative geographic analysis. The digital microscopy scoping review by von Bahr et al. [20] captured one modality subset of the literature mapped here but could not identify cross-modality patterns or the overall disease coverage inequity. This review's finding that only 6.8% of studies report prospective field validation provides a specific, quantified benchmark against which future progress in the field can be measured.

4.9 | Strengths and Limitations

This scoping review has several strengths: a registered protocol with no required amendments, comprehensive search across five databases including engineering literature (IEEE Xplore), validated screening methodology (Cohen's kappa = 0.856, exceeding the pre-specified 90% consistency threshold), enrichment of extracted data using PubMed Central full texts and transparent reporting of all methodological decisions including the search strategy refinement process.

Limitations include the inaccessibility of the Global Index Medicus, which may have resulted in under-identification of LMIC-origin literature, particularly from Latin America (LILACS), Africa (AIM), the Eastern Mediterranean (IMEMR), South-East Asia (IMSEAR) and the Western

Pacific (WPRIM); the restriction to English-language publications, which substantively excludes evidence published in French, Spanish, Portuguese, Mandarin, Russian and other languages used in LMIC health-research literatures; the use of automated screening and extraction tools, which, while validated (Cohen's kappa = 0.856), may have introduced systematic biases in the identification of AI techniques, diseases, or geographic information from unstructured text; and the incomplete extraction of some variables (country at 39%, deployment platform at 43%, validation level at 38%) due to reliance on abstracts for 146 studies without PMC full-text availability. The reporting-completeness sensitivity analysis (Section 3.12.2) shows that disease coverage is partly an artefact of which studies report their data well, with HIV over-represented in the complete-case subset and TB under-represented; this caveat applies throughout Section 3.3.

For this revision a fresh search of the grey-literature sources specified in the original protocol but not searched at submission time (WHO IRIS, FIND publications, PATH publications, the Global Fund evaluations) was conducted on 19 May 2026. The search identified a substantive body of policy and guidance literature relevant to the synthesis—including the WHO 2025 policy statement on computer-aided detection for TB screening [65], the WHO 2021 and 2024 AI ethics and governance documents [78, 79], the FIND Data Standard Framework for AI-based Diagnostics [72] and the FIND/WHO 2024 independent validation platform for CAD-TB products. These are cited in Sections 1, 4.6 and 4.7. The grey-literature search did not identify peer-reviewed primary diagnostic-accuracy, feasibility, or implementation studies absent from the database corpus that would have been eligible for inclusion under the original criteria; consequently, the included-studies set remains at 237. The search was conducted by a single reviewer without co-screening, on then-current organisational publication pages and did not cover non-English grey-literature sources, regional WHO offices, or programmatic reports indexed only in francophone West African or Latin American repositories—limitations that may continue to under-represent regional evidence.

As a scoping review, no formal quality appraisal of included studies was performed, consistent with established JBI methodology [25].

5 | Conclusions

This scoping review maps 237 studies on AI-enhanced point-of-care diagnostics for infectious diseases in resource-limited settings, revealing a field growing rapidly but facing critical challenges in translational maturity, disease equity and geographic representation. The evidence base is concentrated around three diseases (COVID-19, malaria, TB), dominated by convolutional neural networks and overwhelmingly at the pre-deployment stage: only 16 studies (6.8%) report prospective field validation, while 62% do not report their validation level at all.

Six recommendations follow from this review, stratified across three tiers (see Section 4.7 for the empirical basis of each). *Technical*: (T1) match architectural complexity to deployment context, with explicit reporting of the accuracy/inference-cost trade-off—the most-cited heavy-architecture studies (ResNet, EfficientNet, Vision Transformer family) achieve no prospective field validation in this corpus; (T2) build federated learning and privacy-preserving training infrastructure for LMIC POC contexts, addressing a near-total absence of these approaches from the included literature. *Methodological*: (M1) tie funding to pre-registered, multi-site prospective field validation with explicit cost reporting—cost data are reported in only 38% of field-validated studies, a rate indistinguishable from laboratory studies; (M2) adopt a POC-AI-LMIC reporting extension under the EQUATOR Network, with mandatory fields covering target user, hardware envelope, training-data provenance, cost-per-test and failure-mode categorisation. *Systemic*: (S1) establish a WHO prequalification pathway for AI-enhanced diagnostics in parallel with investment in LMIC computational capacity; (S2) issue equity-driven funding calls that ring-fence under-represented disease–region pairs (NTDs, STIs, AMR) and require end-user co-design from the design stage. Equity considerations and the relevance of non-AI rapid-diagnostic alternatives (CRISPR visual readouts, isothermal nucleic-acid amplification, paper-based colourimetric assays) are addressed in Section 4.7.1.

Closing the gap between AI diagnostic promise and real-world impact in resource-limited settings will require a deliberate shift in research priorities: from algorithm benchmarking to field validation, from disease-concentrated to equity-driven research agendas and from HIC-centred development to LMIC-embedded co-design. The tools exist to enhance infectious disease diagnosis at the point of care; the challenge now is to validate and deploy them where they are needed most.

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Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The extracted data, screening decisions and analysis scripts are available in the project repository. The review protocol is registered at https://doi.org/10.17605/OSF.IO/KV8MP.

References

1. WHO, *Essential Diagnostics List* (World Health Organization, 2023).

2. R. W. Peeling, K. K. Holmes, D. Mabey, and A. Ronald, "Rapid Tests for Sexually Transmitted Infections (STIs): The Way Forward," *Sexually Transmitted Infections* 82, no. suppl. 5 (2006): v1–v6.

3. WHO, *Universal Health Coverage* (World Health Organization, 2024).

4. K. A. Fleming, S. Horton, M. L. Wilson, et al., "The Lancet Commission on Diagnostics: Transforming Access to Diagnostics," *Lancet* 398, no. 10315 (2021): 1997–2050.

5. H. H. Leslie, Z. Sun, and M. E. Kruk, "Association Between Infrastructure and Observed Quality of Care in 4 Healthcare Services: A Cross-Sectional Study of 4,300 Facilities in 8 Countries," *PLoS Medicine* 14, no. 12 (2017): e1002464.

6. T. Visser, J. Daily, N. Hotte, et al., *Malaria Rapid Diagnostic Test Performance: Results of WHO Product Testing Rounds 1–8* (World Health Organization, 2018).

7. E. J. Topol, "High-Performance Medicine: The Convergence of Human and Artificial Intelligence," *Nature Medicine* 25, no. 1 (2019): 44–56.

8. P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, "AI in Health and Medicine," *Nature Medicine* 28, no. 1 (2022): 31–38.

9. J. A. Quinn, R. Nakasi, P. K. B. Mugagga, et al., "Deep Convolutional Neural Networks for Microscopy-Based Point of Care Diagnostics," *Proceedings of Machine Learning Research* 56 (2016): 271–281.

10. V. Turbé, C. Herbst, T. Mngomezulu, et al., "Deep Learning of HIV Field-Based Rapid Tests," *Nature Medicine* 27, no. 7 (2021): 1165–1170.

11. Z. Z. Qin, M. S. Sander, B. Rai, et al., "Using Artificial Intelligence to Read Chest Radiographs for Tuberculosis Detection: A Multi-Site Evaluation," *Scientific Reports* 9, no. 1 (2019): 15418.

12. K. A. Stewart, S. M. Navarro, S. Kambala, et al., "Trends in Ultrasound Use in Low and Middle Income Countries: A Systematic Review," *International Journal of MCH AIDS* 9, no. 1 (2020): 103–120.

13. FIND, *AI-Powered Diagnostics* (Foundation for Innovative New Diagnostics, 2025), https://www.finddx.org.

14. PATH, *Advancing AI for Global Health Diagnostics* (PATH, 2024), https://www.path.org.

15. WHO, *WHO Consolidated Guidelines on Tuberculosis. Module 2: Screening—Systematic Screening for Tuberculosis Disease* (World Health Organization, 2021).

16. L. Wynants, B. Van Calster, G. S. Collins, et al., "Prediction Models for Diagnosis and Prognosis of Covid-19: Systematic Review and Critical Appraisal," *BMJ (Clinical Research Ed.)* 369 (2020): m1328.

17. C. J. Kelly, A. Karthikesalingam, M. Suleyman, G. Corrado, and D. King, "Key Challenges for Delivering Clinical Impact With Artificial Intelligence," *BMC Medicine* 17, no. 1 (2019): 195.

18. N. Schwalbe and B. Wahl, "Artificial Intelligence and the Future of Global Health," *Lancet* 395, no. 10236 (2020): 1579–1586.

19. B. Wahl, A. Cossy-Gantner, S. Germann, and N. R. Schwalbe, "Artificial Intelligence (AI) and Global Health: How Can AI Contribute to Health in Resource-Poor Settings?," *BMJ Global Health* 3, no. 4 (2018): e000798.

20. T. von Bahr, A. Suutala, V. Diwan, et al., "AI-Supported Digital Microscopy Diagnostics in Primary Health Care Laboratories: Scoping Review," *JMIR* 28, no. 1 (2026): e78500.

21. M. Harris, A. Qi, L. Jeagal, et al., "A Systematic Review of the Diagnostic Accuracy of Artificial Intelligence-Based Computer Programs to Analyze Chest X-Rays for Pulmonary Tuberculosis," *PLoS One* 14, no. 9 (2019): e0221339.

22. A. Loddo, C. Di Ruberto, and M. Kocher, "Recent Advances of Malaria Parasites Detection Systems Based on Mathematical Morphology," *Sensors* 18, no. 2 (2018): 513.

23. M. D. J. Peters, C. Marnie, A. C. Tricco, et al., "Updated Methodological Guidance for the Conduct of Scoping Reviews," *JBI Evidence Synthesis* 18, no. 10 (2020): 2119–2126.

24. D. Pollock, M. D. J. Peters, H. Khalil, et al., "Recommendations for the Extraction, Analysis, and Presentation of Results in Scoping Reviews," *JBI Evidence Synthesis* 21, no. 3 (2023): 520–532.

25. M. D. J. Peters, C. Godfrey, P. McInerney, et al., "Chapter 11: Scoping Reviews," in *JBI Manual for Evidence Synthesis*, ed. E. Aromataris and Z. Munn (JBI, 2020).

26. A. C. Tricco, E. Lillie, W. Zarin, et al., "PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation," *Annals of Internal Medicine* 169, no. 7 (2018): 467–473.

27. J. R. Landis and G. G. Koch, "The Measurement of Observer Agreement for Categorical Data," *Biometrics* 33, no. 1 (1977): 159–174.

28. Z. Munn, M. D. J. Peters, C. Stern, C. Tufanaru, A. McArthur, and E. Aromataris, "Systematic Review or Scoping Review? Guidance for Authors When Choosing Between a Systematic or Scoping Review Approach," *BMC Medical Research Methodology* 18, no. 1 (2018): 143.

29. T. Xu, Y. Zhang, S. Li, et al., "Deep Learning-Enhanced Hand-Driven Microfluidic Chip for Multiplexed Nucleic Acid Detection Based on RPA/CRISPR," *Advanced Science* 12, no. 21 (2025), https://doi.org/10.1002/advs.202414918.

30. Z. Yang, J. Li, J. Shen, et al., "Recent Progress in Tuberculosis Diagnosis: Insights Into Blood-Based Biomarkers and Emerging Technologies," *Frontiers in Cellular and Infection Microbiology* 15 (2025), https://doi.org/10.3389/fcimb.2025.1567592.

31. F. Bahati, K. O. Owuor, E. Milimo, et al., "Artificial Intelligence (AI)-driven Rapid Diagnostic Test Interpretation in a Connected Diagnostics (ConnDx) System for Dynamic Malaria Surveillance," *BMC Digital Health* 3, no. 1 (2025), https://doi.org/10.1186/s44247-025-00197-x.

32. K. A. Lindblade, C. Ngufor, W. Yavo, et al., "Evaluating the Performance of an Artificial Intelligence-Based Electronic Reader for Malaria Rapid Diagnostic Tests Across Benin, Côte d'ivoire, Nigeria and Uganda," *Malaria Journal* 24, no. 1 (2025), https://doi.org/10.1186/s12936-025-05522-3.

33. S. F. Weber, R. Wolf, K. Manten, et al., "Lung and Abdominal Ultrasound Accuracy for Tuberculosis: An Indian Prospective Cohort Study," *PLoS One* 20, no. 9 (2025): e0329670, https://doi.org/10.1371/journal.pone.0329670.

34. D. Kessler, M. Zhu, C. R. Gregory, et al., "Development and Testing of a Deep Learning Algorithm to Detect Lung Consolidation Among Children With Pneumonia Using Hand-Held Ultrasound," *PLoS One* 19, no. 8 (2024): e0309109, https://doi.org/10.1371/journal.pone.0309109.

35. O. Holmström, S. Stenman, A. Suutala, et al., "A Novel Deep Learning-Based Point-of-Care Diagnostic Method for Detecting Plasmodium Falciparum With Fluorescence Digital Microscopy," *PLoS One* 15, no. 11 (2020): e0242355, https://doi.org/10.1371/journal.pone.0242355.

36. X. Guo, M. A. Khalid, I. Domingos, et al., "Smartphone-Based DNA Malaria Diagnostics Using Deep Learning for Local Decision Support and Blockchain Technology for Security," *Nature Electronics* 4, no. 8 (2021): 615–624, https://doi.org/10.1038/s41928-021-00612-x.

37. V. M. Prathap and S. Yadav, "Enhanced YOLO-Based Framework and Benchmarking for Automated Plasmodium Vivax Detection," *Parasitology Research* 125, no. 1 (2026), https://doi.org/10.1007/s00436-026-08642-0.

38. A. Afifi, N. E. Hafsa, M. A. S. Ali, A. Alhumam, and S. Alsalman, "An Ensemble of Global and Local-Attention Based Convolutional Neural Networks for COVID-19 Diagnosis on Chest X-Ray Images," *Symmetry* 13, no. 1 (2021): 113, https://doi.org/10.3390/sym13010113.

39. S. D. Roche, O. I. Ekwunife, R. Mendonca, et al., "Measuring the Performance of Computer Vision Artificial Intelligence to Interpret

Images of HIV Self-Testing Results," *Frontiers in Public Health* 12 (2024), https://doi.org/10.3389/fpubh.2024.1334881.

40. D. M. Vu, A. R. Krystosik, B. A. Ndenga, et al., "Detection of Acute Dengue Virus Infection, With and Without Concurrent Malaria Infection, in a Cohort of Febrile Children in Kenya, 2014–2019, by Clinicians or Machine Learning Algorithms," *PLOS Global Public Health* 3, no. 7 (2023): e0001950, https://doi.org/10.1371/journal.pgph.0001950.

41. D. Brenes, C. J. Barberan, B. Hunt, et al., "Multi-Task Network for Automated Analysis of High-Resolution Endomicroscopy Images to Detect Cervical Precancer and Cancer," *Computerized Medical Imaging and Graphics* 97 (2022): 102052, https://doi.org/10.1016/j.compmedimag.2022.102052.

42. A. Elbehiry and A. Abalkhail, "Metagenomic Next-Generation Sequencing in Infectious Diseases: Clinical Applications, Translational Challenges, and Future Directions," *Diagnostics* 15, no. 16 (2025): 1991, https://doi.org/10.3390/diagnostics15161991.

43. J. R. Zech, M. A. Badgeley, M. Liu, A. B. Costa, J. J. Titano, and E. K. Oermann, "Variable Generalization Performance of a Deep Learning Model to Detect Pneumonia in Chest Radiographs: A Cross-Sectional Study," *PLoS Medicine* 15, no. 11 (2018): e1002683.

44. C. Ho, C. Puthur, B. Nabatte, et al., "Interpretable Machine Learning and Signal Processing for Automated Reading and Quality Control of Lateral Flow Tests for Schistosomiasis," medRxiv Preprint, 2025, https://doi.org/10.1101/2025.10.01.25337079.

45. M. Mahajan, A. S. Kalra, A. Bansal, A. Bhattacherjee, Vishal, and E. Jain, "Automated Malaria Detection Using Deep Learning: A Convolutional Neural Network Approach for Microscopic Cell Image Analysis," in *2024 International Conference on Decision Aid Sciences and Applications (DASA)* (2024), 1–6, https://doi.org/10.1109/DASA63652.2024.10836383.

46. Y. Zhong, Y. Dan, Y. Cai, et al., "Efficient Malaria Parasite Detection From Diverse Images of Thick Blood Smears for Cross-Regional Model Accuracy," *IEEE Open Journal of Engineering in Medicine and Biology* 4 (2023): 226–233, https://doi.org/10.1109/OJEMB.2023.3328435.

47. R. Parveen, B. Qui, W. Song, et al., "Trustworthy Deep Learning for Malaria Diagnosis Using Explainable Artificial Intelligence," *Scientific Reports* 15, no. 1 (2025), https://doi.org/10.1038/s41598-025-28387-7.

48. L. Oktovianus, J. K. Tikkhaviro, and S. Y. Prasetyo, "Benchmarking CNN Models for Malaria Cell Detection," in *2024 International Conference on Computer Engineering, Network, and Intelligent Multimedia (CENIM)* (2024), 1–6, https://doi.org/10.1109/CENIM64038.2024.10882645.

49. S. Sharma, R. Pradhan, and A. Varshney, "Deep Learning for Malaria Detection: A CNNBased Classification of Blood Smear Images," in *2025 5th Asian Conference on Innovation in Technology (ASIANCON)* (2025), 1–6, https://doi.org/10.1109/ASIANCON66527.2025.11280660.

50. C. Nithisha, K. Kushitha, K. Revanth, S. V. S. N. Reddy, A. P. Sree, and R. S. Teja, "Enhancing Malaria Detection From Blood Smear Images Using Hybrid Deep Learning Models," in *2025 4th OPJU International Technology Conference (OTCON) on Smart Computing for Innovation and Advancement in Industry 5.0* (2025), 1–5, https://doi.org/10.1109/OTCON65728.2025.11070759.

51. F. Kone, L. Conrad, J. T. Coulibaly, et al., "MALDI-TOF Mass Spectrometry Combined With Machine Learning Algorithms to Identify Protein Profiles Related to Malaria Infection in Human Sera From Côte d'ivoire," *Malaria Journal* 24, no. 1 (2025), https://doi.org/10.1186/s12936-025-05362-1.

52. S. Chattoraj, B. Reddy, M. Tadepalli, and P. Putha, "Comparing Deep Learning Models for Tuberculosis Detection: A Retrospective Study of Digital vs. Analog Chest Radiographs," *Indian Journal of Tuberculosis* 72 (2025): S43–S46, https://doi.org/10.1016/j.ijtb.2024.05.008.

53. N. Marquez, E. J. Carpio, M. R. Santiago, et al., "Performance of Chest X-Ray With Computer-Aided Detection Powered by Deep Learning-Based Artificial Intelligence for Tuberculosis Presumptive Identification During Case Finding in the Philippines," *BMC Global and Public Health* 3, no. 1 (2025), https://doi.org/10.1186/s44263-025-00198-y.

54. S. Khan, F. Siddiqui, and M. A. Ahad, "Integrating CNN-Based Feature Fusion and Spatial Attention for Tuberculosis Detection With Bayesian Optimized XGBoost," *International Journal of Data Science and Analytics* 20, no. 6 (2025): 5415–5434, https://doi.org/10.1007/s41060-025-00778-z.

55. H. Sagreiya, M. A. Jacobs, and A. Akhbardeh, "Automated Lung Ultrasound Pulmonary Disease Quantification Using an Unsupervised Machine Learning Technique for COVID-19," *Diagnostics* 13, no. 16 (2023): 2692, https://doi.org/10.3390/diagnostics13162692.

56. M. A. Rohaim, E. Clayton, I. Sahin, et al., "Artificial Intelligence-Assisted Loop Mediated Isothermal Amplification (AI-LAMP) for Rapid Detection of SARS-CoV-2," *Viruses* 12, no. 9 (2020): 972, https://doi.org/10.3390/v12090972.

57. M. Chandra, K. Kumar, P. Thakur, S. Chattopadhyaya, F. Alam, and S. Kumar, "Digital Technologies, Healthcare and Covid-19: Insights From Developing and Emerging Nations," *Health and Technology* 12, no. 2 (2022): 547–568, https://doi.org/10.1007/s12553-022-00650-1.

58. O. Holmström, N. Linder, H. Kaingu, et al., "Point-of-Care Digital Cytology With Artificial Intelligence for Cervical Cancer Screening in a Resource-Limited Setting," *JAMA Network Open* 4, no. 3 (2021): e211740, https://doi.org/10.1001/jamanetworkopen.2021.1740.

59. D. Bermejo-Peláez, N. Medina, E. Álamo, et al., "Digital Platform for Automatic Qualitative and Quantitative Reading of a Cryptococcal Antigen Point-of-Care Assay Leveraging Smartphones and Artificial Intelligence," *Journal of Fungi* 9, no. 2 (2023): 217, https://doi.org/10.3390/jof9020217.

60. S. Mertens, J. Krois, A. Graf, et al., "Artificial Intelligence and COVID-19: Rapid Review," *BMJ Open* 12, no. 3 (2022): e054511.

61. WHO, *Neglected Tropical Diseases* (World Health Organization, 2025).

62. J. A. Røttingen, S. Regmi, M. Eide, et al., "Mapping of Available Health Research and Development Data: What's There, What's Missing, and What Role Is There for a Global Observatory?," *Lancet* 382, no. 9900 (2013): 1286–1307.

63. E. Beede, E. Baylor, F. Hersch, et al., "A Human-Centered Evaluation of a Deep Learning System Deployed in Clinics for the Detection of Diabetic Retinopathy," in *Proc CHI* (ACM, 2020), 1–12.

64. L. Seyyed-Kalantari, H. Zhang, M. B. A. McDermott, I. Y. Chen, and M. Ghassemi, "Underdiagnosis Bias of Artificial Intelligence Algorithms Applied to Chest Radiographs in Under-Served Patient Populations," *Nature Medicine* 27, no. 12 (2021): 2176–2182.

65. WHO, *Use of Computer-Aided Detection Software for Tuberculosis Screening: WHO Policy Statement* (World Health Organization, 2025), https://www.who.int/publications/i/item/9789240110373.

66. L.-T. Allan-Blitz, G. Adams, G. Sanders, et al., "Preliminary Clinical Performance of a Cas13a-Based Lateral Flow Assay for Detecting Neisseria Gonorrhoeae in Urine Specimens," *mSphere* 10, no. 1 (2025), https://doi.org/10.1128/msphere.00677-24.

67. D. Hou, R. Hou, and J. Hou, "Interpretable Saab Subspace Network for COVID-19 Lung Ultrasound Screening," in *2020 11th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)* (2020), 0393–0398, https://doi.org/10.1109/UEMCON51285.2020.9298069.

68. A. Colubri, M. A. Hartley, M. Siakor, et al., "Machine-Learning Prognostic Models From the 2014-16 Ebola Outbreak: Data-Harmonization Challenges, Validation Strategies, and mHealth Applications," *EClinicalMedicine* 11 (2019): 54–64, https://doi.org/10.1016/j.eclinm.2019.06.003.

69. M. Pai, S. G. Schumacher, and S. Abimbola, "Surrogate Endpoints in Global Health Research: Still Searching for Killer Apps and Silver Bullets?," *BMJ Global Health* 3, no. 2 (2018): e000755.

70. N. Rieke, J. Hancox, W. Li, et al., "The Future of Digital Health With Federated Learning," *npj Digital Medicine* 3 (2020): 119.

71. M. Raghu, C. Zhang, J. Kleinberg, and S. Bengio, "Transfusion: Understanding Transfer Learning for Medical Imaging," in *Proc NeurIPS*, vol. 32 (ACM, 2019).

72. Foundation for Innovative New Diagnostics (FIND), *Data Standard Framework for AI-Based Diagnostics* (FIND, 2024), https://www.finddx.org/what-we-do/cross-cutting-workstreams/digital-health/data-standard-framework-for-ai-based-diagnostics/.

73. The Global Fund to Fight AIDS, Tuberculosis and Malaria, *Global Fund and Siemens Healthineers to Accelerate Adoption of Artificial Intelligence for Tuberculosis Diagnosis. Press Release, 16 October 2023* (Global Fund, 2023), https://www.theglobalfund.org/en/news/2023/2023-10-16-global-fund-siemens-healthineers-accelerate-adoption-artificial-intelligence-tuberculosis-diagnosis/.

74. V. Sounderajah, H. Ashrafian, R. Aggarwal, et al., "Developing Specific Reporting Guidelines for Diagnostic Accuracy Studies Assessing AI Interventions: The STARD-AI Steering Group," *Nature Medicine* 26, no. 6 (2020): 807–808.

75. G. S. Collins and K. G. M. Moons, "Reporting of Artificial Intelligence Prediction Models," *Lancet* 393, no. 10181 (2019): 1577–1579.

76. I. Simera, D. Moher, A. Hirst, J. Hoey, K. F. Schulz, and D. G. Altman, "Transparent and Accurate Reporting Increases Reliability, Utility, and Impact of Your Research: Reporting Guidelines and the EQUATOR Network," *BMC Medicine* 8 (2010): 24.

77. WHO, *WHO Prequalification of In Vitro Diagnostics* (World Health Organization, 2025), https://www.who.int/teams/regulation-prequalification/in-vitro-diagnostics.

78. WHO, *Ethics and Governance of Artificial Intelligence for Health: Guidance on Large Multi-Modal Models* (World Health Organization, 2024), https://iris.who.int/handle/10665/375579.

79. WHO, *Ethics and Governance of Artificial Intelligence for Health: WHO Guidance* (World Health Organization, 2021), https://iris.who.int/handle/10665/341996.

80. Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations," *Science* 366, no. 6464 (2019): 447–453.

81. J. S. Gootenberg, O. O. Abudayyeh, J. W. Lee, et al., "Nucleic Acid Detection With CRISPR-Cas13a/C2c2," *Science* 356, no. 6336 (2017): 438–442.

82. J. S. Chen, E. Ma, L. B. Harrington, et al., "CRISPR-Cas12a Target Binding Unleashes Indiscriminate Single-Stranded DNase Activity," *Science* 360, no. 6387 (2018): 436–439.

83. C. Myhrvold, C. A. Freije, J. S. Gootenberg, et al., "Field-Deployable Viral Diagnostics Using CRISPR-Cas13," *Science* 360, no. 6387 (2018): 444–448.

84. P. Craw and W. Balachandran, "Isothermal Nucleic Acid Amplification Technologies for Point-of-Care Diagnostics: A Critical Review," *Lab on a Chip* 12, no. 14 (2012): 2469–2486.

85. O. Piepenburg, C. H. Williams, D. L. Stemple, and N. A. Armes, "DNA Detection Using Recombination Proteins," *PLoS Biology* 4, no. 7 (2006): e204.

86. A. W. Martinez, S. T. Phillips, M. J. Butte, and G. M. Whitesides, "Patterned Paper as a Platform for Inexpensive, Low-Volume, Portable Bioassays," *Angewandte Chemie (International Ed. in English)* 46, no. 8 (2007): 1318–1320.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Formal search strategies—all databases. **Data S2:** PRISMA-ScR checklist. **Data S3:** PRISMA-ScR flow diagram. **Data S4:** Grey-literature search log (May 2026). **Data S5:** SuppInfo_5_analysis_outputs. **Table S1:** Complete data charting table. **Table S2:** Excluded studies with reasons.